Data Analysis on Black Friday sales in a retail store using Machine Learning Models and Vector Support Techniques

Data Science: Capstone Project for Harvardx Professional Data Science Certificate
(Choose your own project: PH125.9x)

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1 Overview

The purpose of this project is to apply advanced machine learning techniques and support vector machine techniques to a publicly available dataset [1] as part of the 'Data Science: Capstone' course (PH125.9x) offered by edX HarvardX. The project goes beyond traditional linear regression methods and focuses on exploring and analyzing the data using various machine learning algorithms. The goal is to gain insights from the analysis and effectively communicate the process and findings.

1.1 Introduction

Black Friday is a major shopping event that takes place on the day after Thanksgiving Day in the United States. It is typically regarded as the beginning of the Christmas shopping season and is characterized by large discounts and promotions offered by retailers. Black Friday is considered one of the busiest shopping days of the year, with crowds of shoppers lining up outside stores in the early morning hours to take advantage of the deals. In recent years, the popularity of online shopping has led to the emergence of Cyber Monday, which takes place on the Monday after Black Friday and features online sales and discounts. Black Friday is now also observed in other parts of the world, including Canada, the United Kingdom, and Australia.

1.2 Project Description

The objective of this project is to use machine learning models and vector support techniques to evaluate and predict the Black Friday sales volume with different option as Gender, Top Sellers, Age, City, Marital Status and occupation. Data transformation and feature engineering are included to improve the predictions and, as various modeling approaches are presented, Several modeling techniques will be explored and the most suitable one will be identified based on evaluation metrics such as Root Mean Square Error (RMSE).

1.3 Data Set

The present project studies the Black Friday Dataset, available on the Kaggle [1] community website. This dataset consists of 12 different columns with 537,577 records, each representing following corresponding variables:

1. User_ID: Unique identifier of shopper.
2. Product_ID: Unique identifier of product. (No key given)
3. Gender: Sex of shopper.
4. Age: Age of shopper split into bins.
5. Occupation: Occupation of shopper. (No key given)
6. City_Category: Residence location of shopper. (No key given)
7. Stay_In_Current_City_Years: Number of years stay in current city.
8. Marital_Status: Marital status of shopper.
9. Product_Category_1: Product category of purchase.
10. Product_Category_2: Product may belong to other category.
11. Product_Category_3: Product may belong to other category.
12. Purchase: Purchase amount in dollars.

The upcoming section will include data exploration, visualization, transformation, and feature engineering on the referenced dataset.

1.4 Attribute Statistics

Basic statistics about the data are obtained in the below:

```
##     User_ID         Product_ID           Gender              Age
##  Min.   :1000001   Length:537577      Length:537577      Length:537577
##  1st Qu.:1001495   Class :character   Class :character   Class :character
##  Median :1003031   Mode  :character   Mode  :character   Mode  :character
##  Mean   :1002992
##  3rd Qu.:1004417
##  Max.   :1006040
##
##    Occupation     City_Category      Stay_In_Current_City_Years
##  Min.   : 0.000   Length:537577      Length:537577
##  1st Qu.: 2.000   Class :character   Class :character
##  Median : 7.000   Mode  :character   Mode  :character
##  Mean   : 8.083
##  3rd Qu.:14.000
##  Max.   :20.000
##
##  Marital_Status   Product_Category_1 Product_Category_2 Product_Category_3
##  Min.   :0.0000   Min.   : 1.000     Min.   : 2.00      Min.   : 3.0
##  1st Qu.:0.0000   1st Qu.: 1.000     1st Qu.: 5.00      1st Qu.: 9.0
##  Median :0.0000   Median : 5.000     Median : 9.00      Median :14.0
##  Mean   :0.4088   Mean   : 5.296     Mean   : 9.84      Mean   :12.7
##  3rd Qu.:1.0000   3rd Qu.: 8.000     3rd Qu.:15.00      3rd Qu.:16.0
##  Max.   :1.0000   Max.   :18.000     Max.   :18.00      Max.   :18.0
##                                      NA's   :166986     NA's   :373299
##     Purchase
##  Min.   :  185
##  1st Qu.: 5866
##  Median : 8062
##  Mean   : 9334
##  3rd Qu.:12073
##  Max.   :23961
##
```

```
##   User_ID Product_ID Gender   Age Occupation City_Category
## 1 1000001  P00069042      F  0-17         10             A
## 2 1000001  P00248942      F  0-17         10             A
## 3 1000001  P00087842      F  0-17         10             A
## 4 1000001  P00085442      F  0-17         10             A
## 5 1000002  P00285442      M   55+         16             C
## 6 1000003  P00193542      M 26-35         15             A
##   Stay_In_Current_City_Years Marital_Status Product_Category_1
## 1                          2              0                  3
## 2                          2              0                  1
## 3                          2              0                 12
## 4                          2              0                 12
## 5                         4+              0                  8
## 6                          3              0                  1
##   Product_Category_2 Product_Category_3 Purchase
## 1                 NA                 NA     8370
## 2                  6                 14    15200
```

```
## 3                  NA                  NA      1422
## 4                  14                  NA      1057
## 5                  NA                  NA      7969
## 6                   2                  NA     15227
```

2 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is a crucial step in analyzing a dataset. It involves conducting preliminary investigations on the data to identify patterns, anomalies, and relationships, as well as testing hypotheses and verifying assumptions. This is typically achieved through the use of summary statistics and graphical representations. By performing EDA, analysts can gain a deeper understanding of the data and make informed decisions about how to proceed with subsequent analyses [2].

Upon analyzing the dataset, it becomes clear that each row pertains to a unique transaction and the item purchased by a particular customer. This information is crucial when we group transactions by User_ID to calculate the total purchases made by each customer.

2.1 Gender

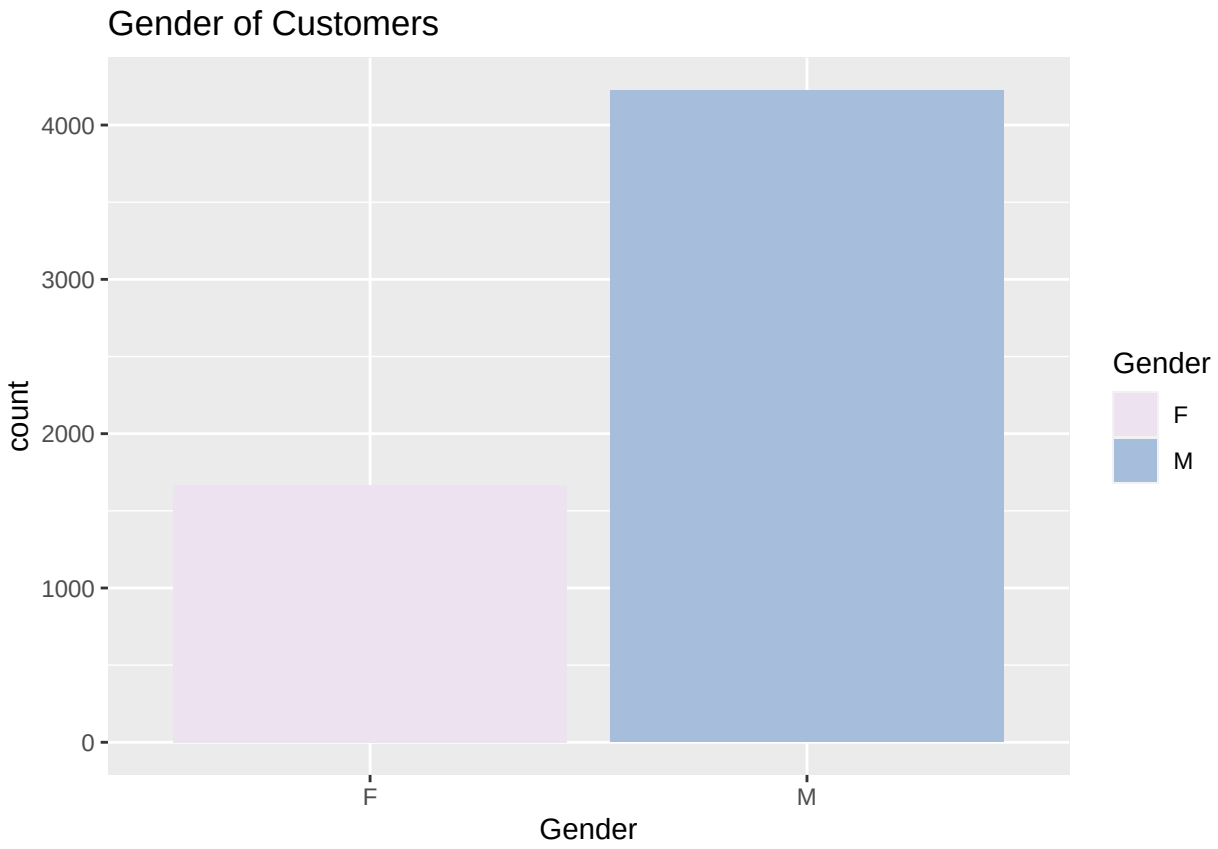
To start analysis, will look at the gender distribution of shoppers in this store. However, since each row in the data represents a unique transaction, need to group the data by User_ID to eliminate any duplicates before can proceed with analysis.

```
## # A tibble: 6 x 2
## # Groups:   User_ID [6]
##   User_ID Gender
##     <int> <chr>
## 1 1000001 F
## 2 1000002 M
## 3 1000003 M
## 4 1000004 M
## 5 1000005 M
## 6 1000006 F
```

```
##    Length     Class      Mode
##      5891 character character
```

```
## # A tibble: 2 x 2
##   Gender n_users
##   <chr>    <int>
## 1 F         1666
## 2 M         4225
```

After created a dataframe that shows each User_ID along with their corresponding gender and transaction counts, we can proceed with visualizing the distribution of gender in our dataset. Follows is plot gender distribution to gain further insights.



The gender split data can be beneficial for retailers as it enables them to make adjustments to their store design, product offerings, and other factors based on the proportion of male and female shoppers.

1. Gender split on Black Friday at our store shows more male shoppers than female shoppers.
2. This information is useful for retailers as it can inform modifications to store layout, product selection, and other variables based on the gender proportion of their shoppers.

To further explore this topic, we can calculate the mean spending amount based on gender. To make the data easier to interpret and analyze, we will create separate tables and then merge them. This will help us identify any differences in spending patterns between men and women, which can inform retailers on how to better cater to their target customers.

```
## # A tibble: 6 x 2
## # Groups:   User_ID [6]
##   User_ID Gender
##     <int> <chr>
## 1 1000001 F
## 2 1000002 M
## 3 1000003 M
## 4 1000004 M
## 5 1000005 M
## 6 1000006 F
```

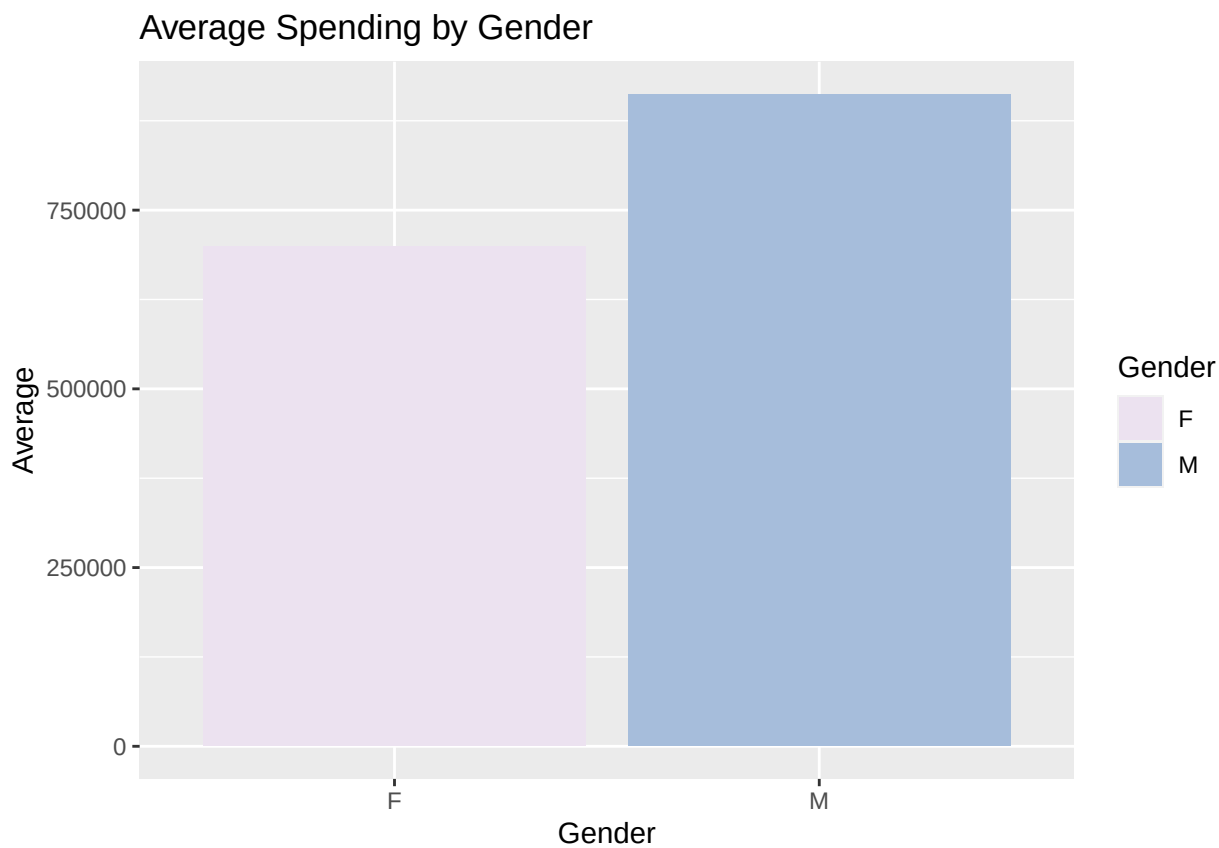
```
## # A tibble: 6 x 2
##   User_ID Total_Purchase
```

```
##     <int>          <int>
## 1 1000001         333481
## 2 1000002         810353
## 3 1000003         341635
## 4 1000004         205987
## 5 1000005         821001
## 6 1000006         379450
```

```
## # A tibble: 6 x 3
##   User_ID Total_Purchase Gender
##     <int>          <int> <chr>
## 1 1000001         333481 F
## 2 1000002         810353 M
## 3 1000003         341635 M
## 4 1000004         205987 M
## 5 1000005         821001 M
## 6 1000006         379450 F
```

Based on our analysis, it appears that the average transaction amount for females is 699054.00, while the average transaction amount for males is 911963.20. To better communicate our findings, we can create visualizations such as charts or graphs. This will allow us to present our results in a clear and concise manner, making it easier for stakeholders to understand the data and make informed decisions.

```
## # A tibble: 2 x 4
##   Gender   Purchase Count Average
##   <chr>       <dbl> <int>   <dbl>
## 1 F      1164624021  1666 699054.
## 2 M      3853044357  4225 911963.
```



Our analysis reveals an interesting trend: while female shoppers make fewer purchases than males at this particular store, their average spending amount is nearly equal to that of male shoppers. However, it is important to note that this observation must be considered in the context of scale, as females are still spending an average of approximately 250,000 less than males. Retailers should take this information into account when developing marketing strategies and product offerings that cater to their diverse customer base.

2.2 Top Seller

Let's now shift our focus to the top-selling products. In this scenario, we won't group the data by product ID since we want to identify any duplicate purchases, in case customers are buying two or more quantities of the same product. This will give us a better understanding of which products are most popular among our customers and help us make informed decisions about inventory management and product promotion.

```
##   Product_ID    n
## 1  P00265242 1858
## 2  P00110742 1591
## 3  P00025442 1586
## 4  P00112142 1539
## 5  P00057642 1430
```

Based on our analysis, it appears that the top five best-selling products, ranked by product ID and corresponding quantity sold, are:

1. P00265242 = 1858

2. P00110742 = 1591
3. P00025442 = 1586
4. P00112142 = 1539
5. P00057642 = 1430

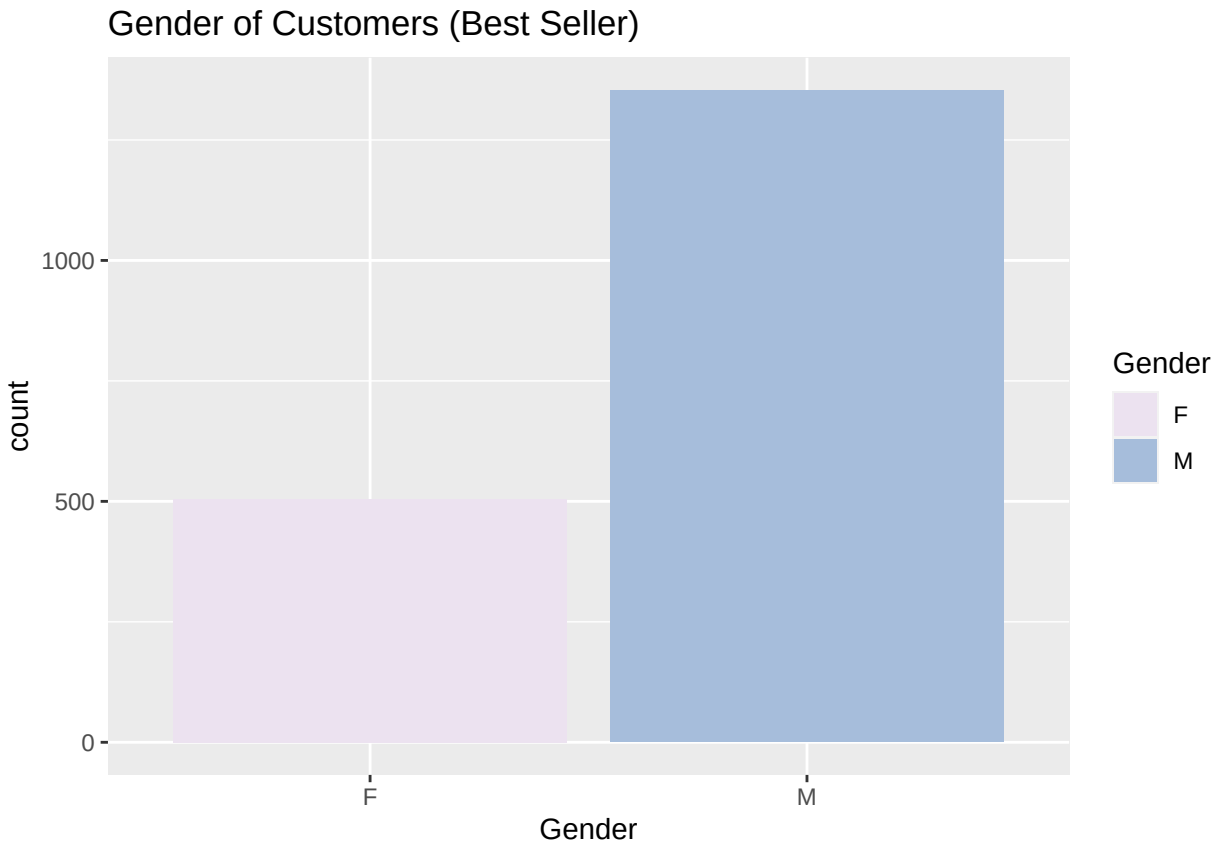
Now that we have identified the top-selling products, let's take a closer look at the best-selling product, P00265242. This analysis can help us understand what makes this product so popular among our customers and help us identify opportunities to increase sales and revenue.

```
##      User_ID Product_ID Gender   Age Occupation City_Category
## 400  1000066  P00265242      M 26-35         18             C
## 1192 1000196  P00265242      F 36-45          9             C
## 1373 1000222  P00265242      M 26-35          1             A
## 1846 1000301  P00265242      M 18-25          4             B
## 2210 1000345  P00265242      M 26-35         12             A
## 2405 1000383  P00265242      F 26-35          7             A
##      Stay_In_Current_City_Years Marital_Status Product_Category_1
## 400                           2              0                  5
## 1192                         4+              0                  5
## 1373                          1              0                  5
## 1846                         4+              0                  5
## 2210                          2              1                  5
## 2405                         4+              1                  5
##      Product_Category_2 Product_Category_3 Purchase
## 400                   8                 NA     8652
## 1192                  8                 NA     8767
## 1373                  8                 NA     6944
## 1846                  8                 NA     8628
## 2210                  8                 NA     8593
## 2405                  8                 NA     6998
```

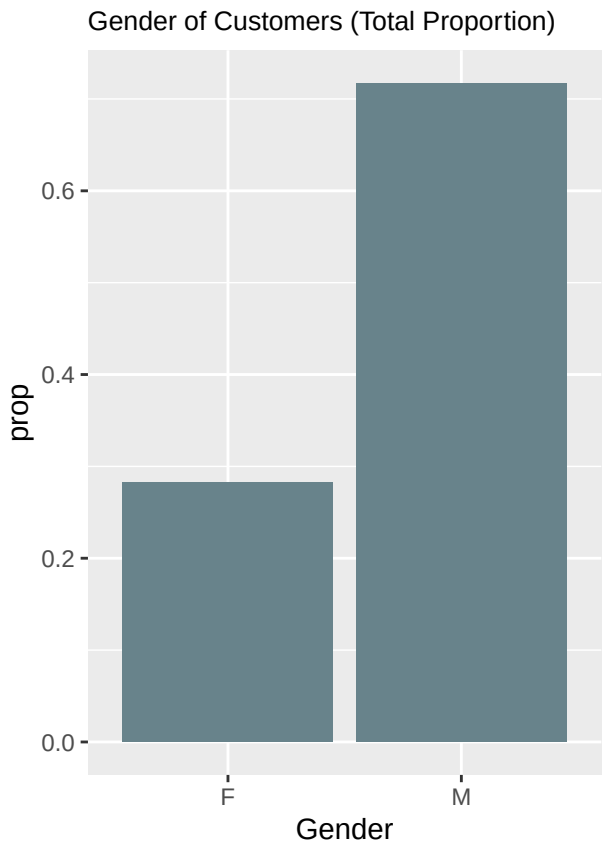
Our analysis reveals that the best-selling product, P00265242, falls under Product_Category_1 = 5 and Product_Category_2 = 8. However, in order to determine the specific name of the product, we would need a key or reference guide.

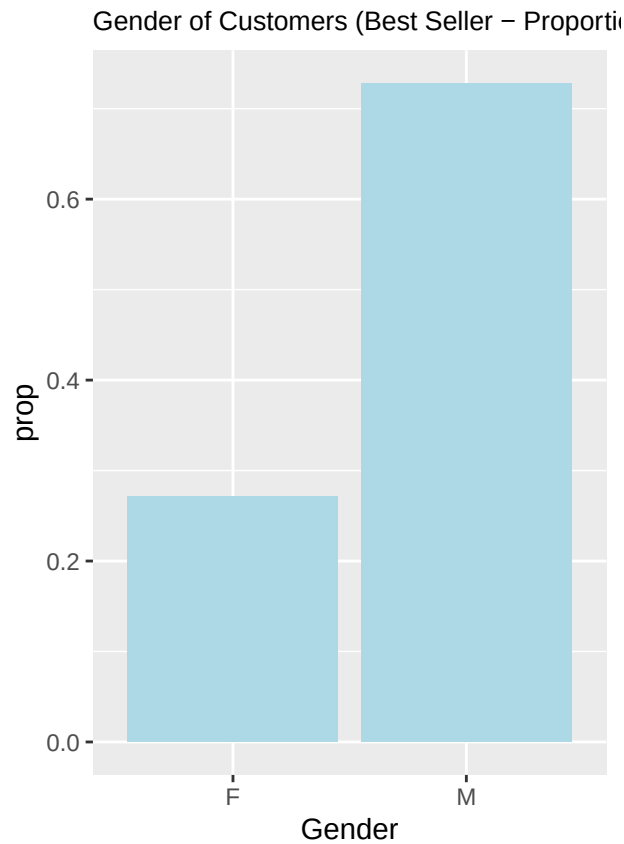
Another interesting observation is that customers are purchasing the same product at different prices. This could be due to various Black Friday promotions, discounts, or coupon codes. Alternatively, there may be other factors at play that are causing price discrepancies between customers, which would require further investigation.

Now, let's examine whether any gender-related patterns exist in the sales of this best-selling product. This analysis can help us identify any differences in purchasing behavior between men and women and help us tailor our marketing strategies accordingly.



We observe a comparable gender distribution in the subset of data as in the overall dataset. To verify this, let's perform a statistical test.





The gender distribution among purchasers of the bestseller and all products appears to be roughly 25% female and 75% male in the overall observation set. While a slight difference exists, it is reasonable to conclude that the bestseller does not target a specific gender.

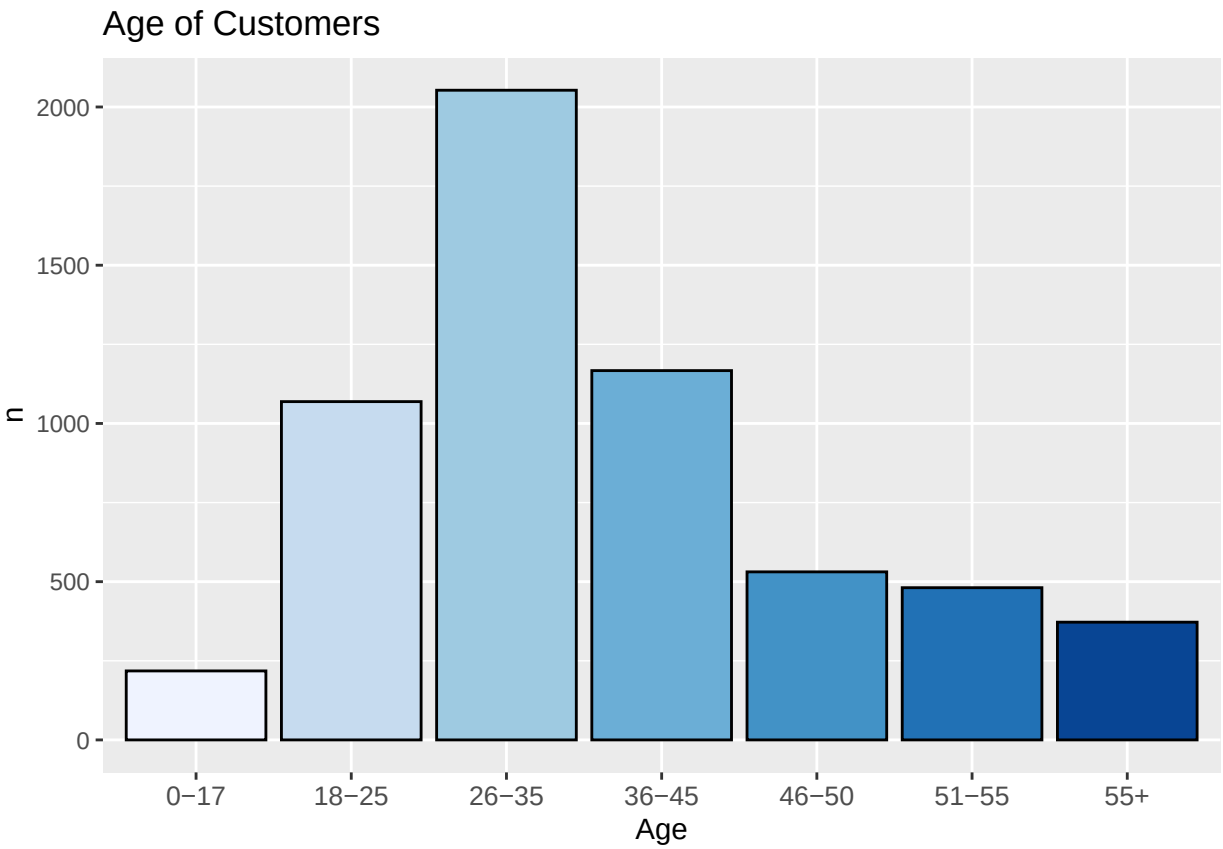
Next, let's shift our focus to analyzing the age variable.

2.3 Age

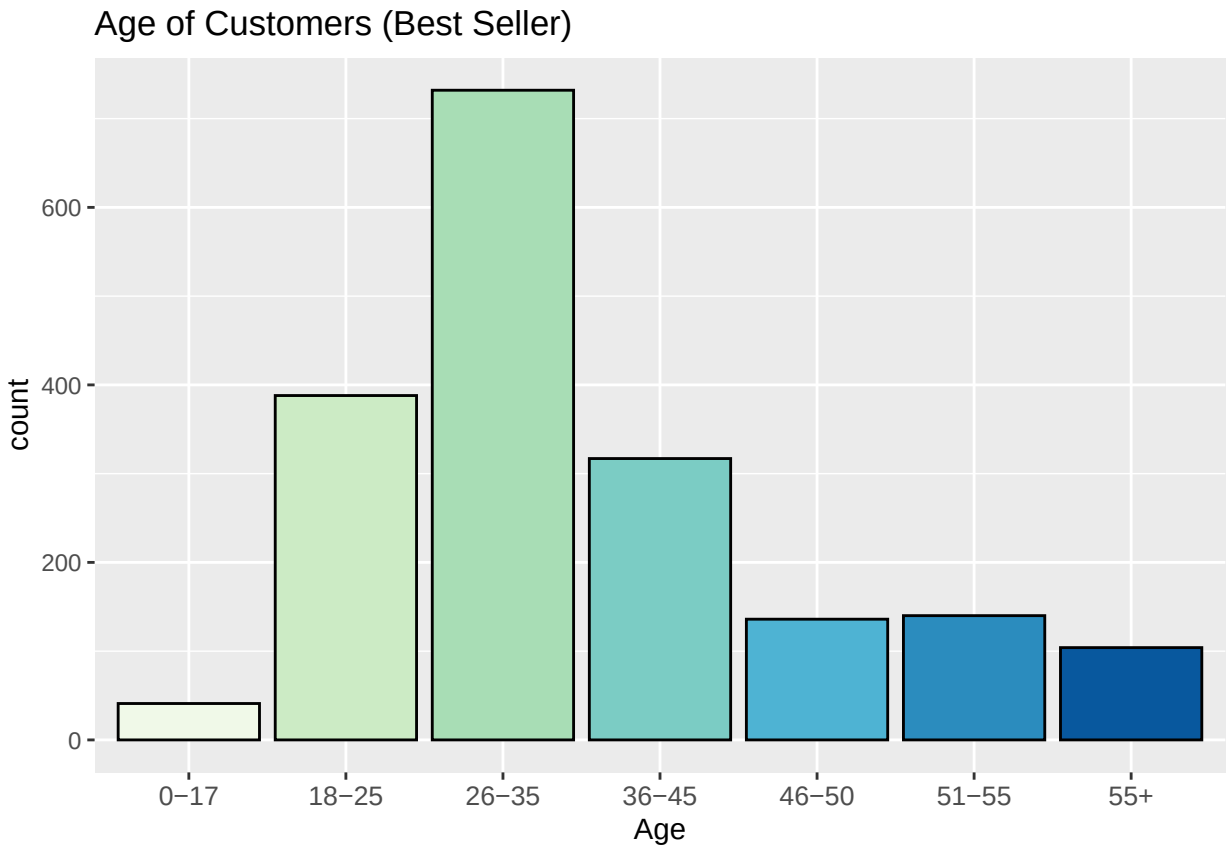
We can start our analysis of age by generating a table that shows the number of individuals in each age bracket.

```
##     Age    n
## 1  0-17  218
## 2 18-25 1069
## 3 26-35 2053
## 4 36-45 1167
## 5 46-50  531
## 6 51-55  481
## 7   55+  372
```

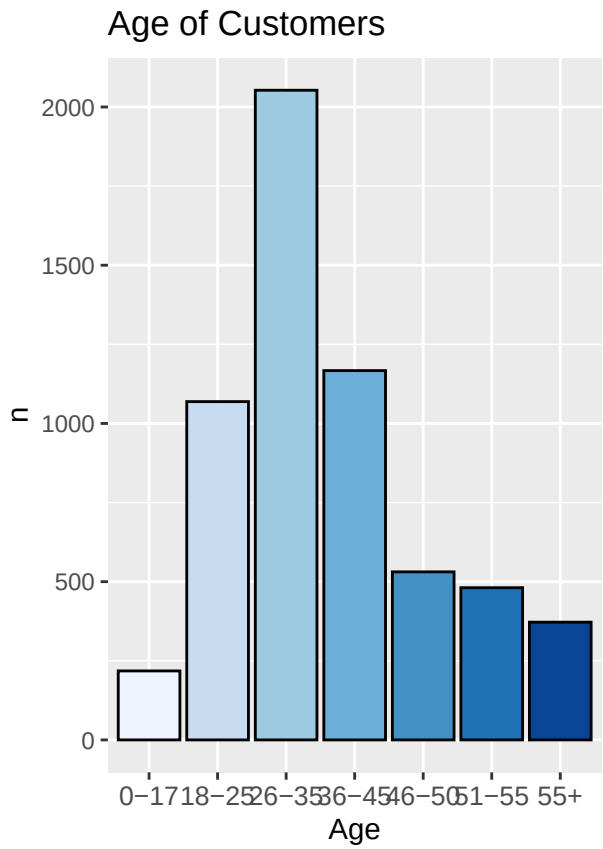
Our dataset includes the counts of customers in each age group who visit our store. It would be beneficial to visualize this data in a graphical format.

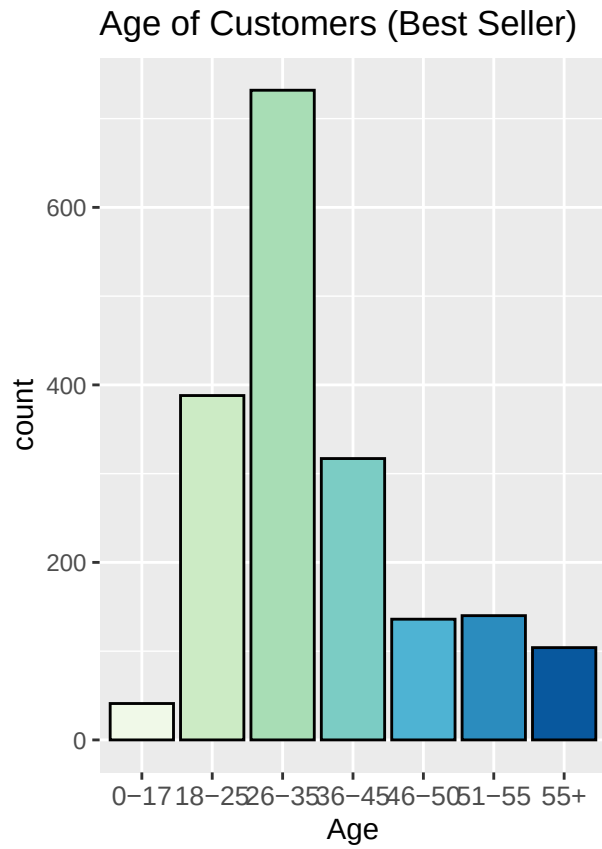


Another chart that we could create is one that illustrates the age distribution of our "best seller" category. This would enable us to identify whether there is a particular age group that has a higher tendency to purchase our best-selling product compared to other shoppers.



Based on our observations, it appears that individuals in the age groups of 18-25 and 26-35 have made the highest number of purchases of our best-selling product. We should now compare this finding with the data from the overall dataset to determine if this trend holds true across all purchases.





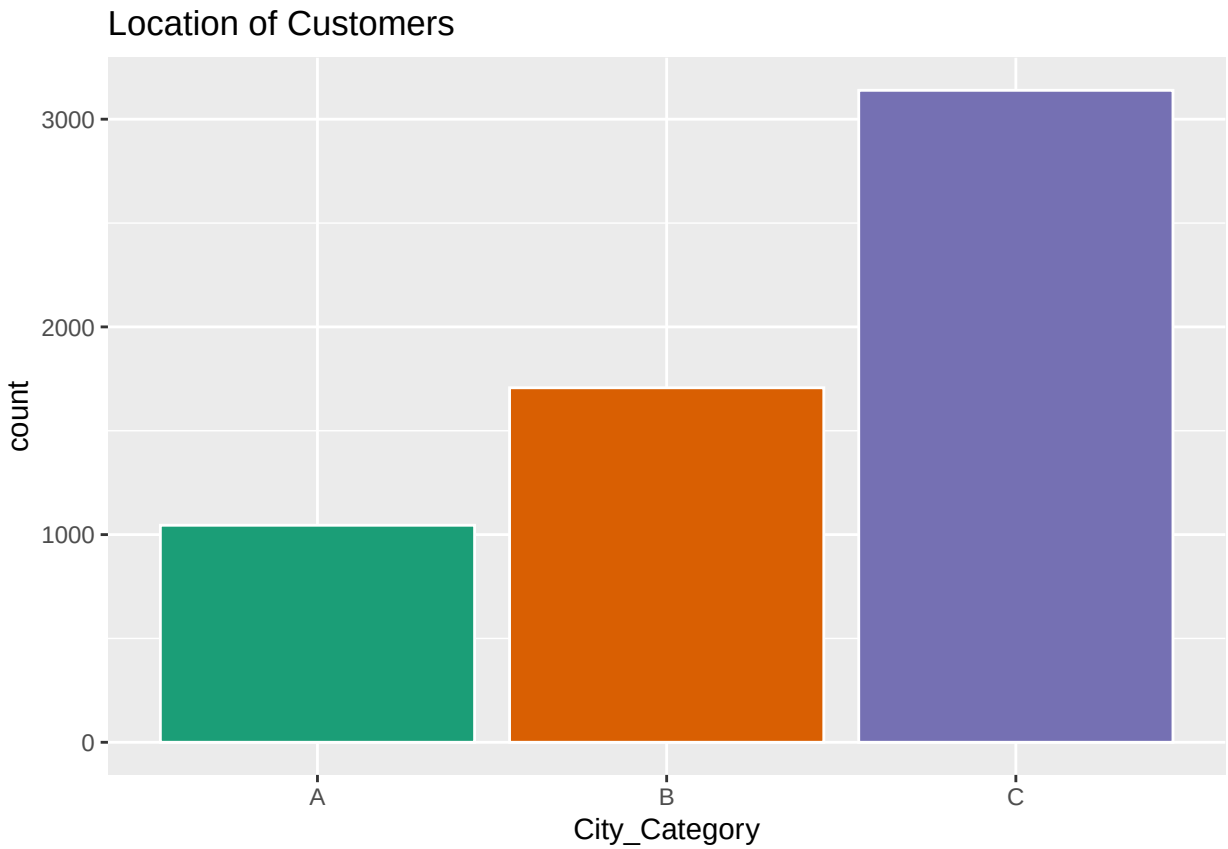
Upon comparing the proportion of customers by age between the best-selling product and the overall dataset, we can observe some variation. Specifically, it seems that customers who are aged 45 and above purchase the top-selling product slightly less frequently than other products in the overall dataset.

Having analyzed the variable of age, we can now shift our focus to another variable.

2.4 City

Let's create a table that shows each User_ID and their respective City_Category.

```
##   User_ID City_Category
## 1 1000001             A
## 2 1000002             C
## 3 1000003             A
## 4 1000004             B
## 5 1000005             A
## 6 1000006             A
```



After analyzing the data, it's evident that the majority of our customers reside in City C. Next, we can calculate the total purchase amount made by customers in each city to determine which city's customers have spent the highest amount at our store.

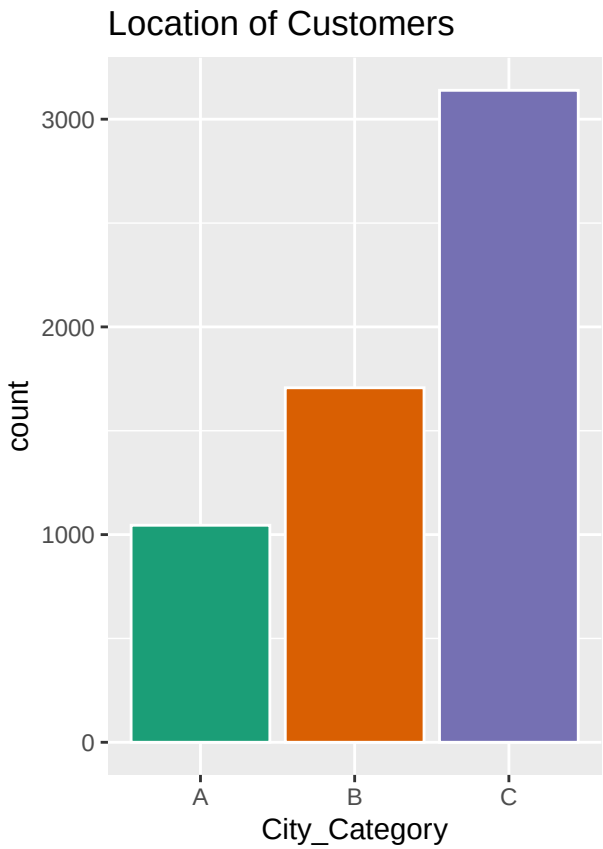
```
## # A tibble: 3 x 3
##   City_Category  Purchases purchasesThousands
##   <chr>              <int>              <dbl>
## 1 A             1295668797           1295669.
## 2 B             2083431612           2083432.
## 3 C             1638567969           1638568.
```

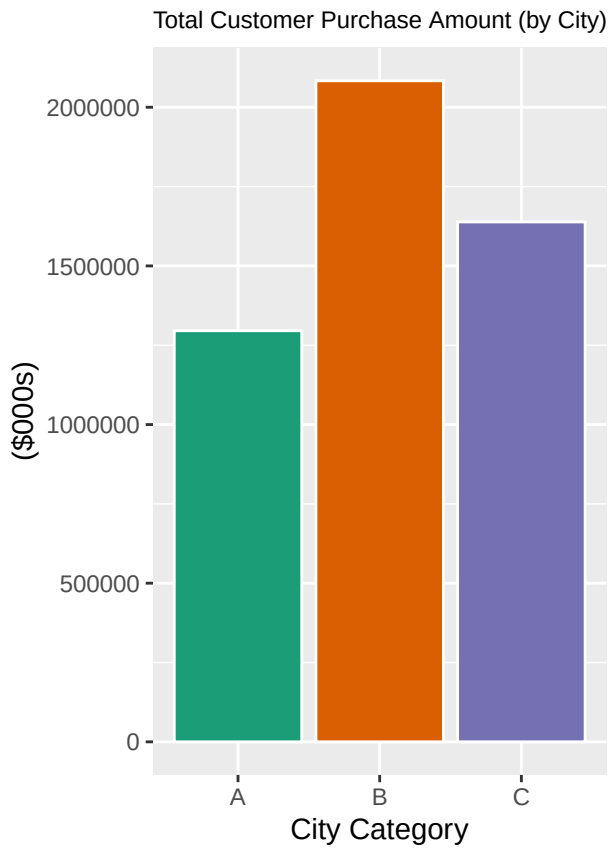
To make the numbers more manageable, we divided the Purchases column by 1000. This is a common practice in the business and accounting world, as it simplifies large numbers and makes them easier to read and plot on charts.

With our table in place, it's time to create a visual representation of our findings.

To make the numbers more manageable, we divided the Purchases column by 1000. This is a common practice in the business and accounting world, as it simplifies large numbers and makes them easier to read and plot on charts.

With our table in place, it's time to create a visual representation of our findings.





From the chart, it's clear that City C had the most number of customers who shopped at our store on Black Friday. However, customers from City B had the highest total purchase amount.

To gain a deeper understanding of this observation, we need to conduct further analysis. Our next step is to determine the number of purchases made by customers in each city. To do so, we will first need to calculate the total number of purchases for each User_ID in the dataset.

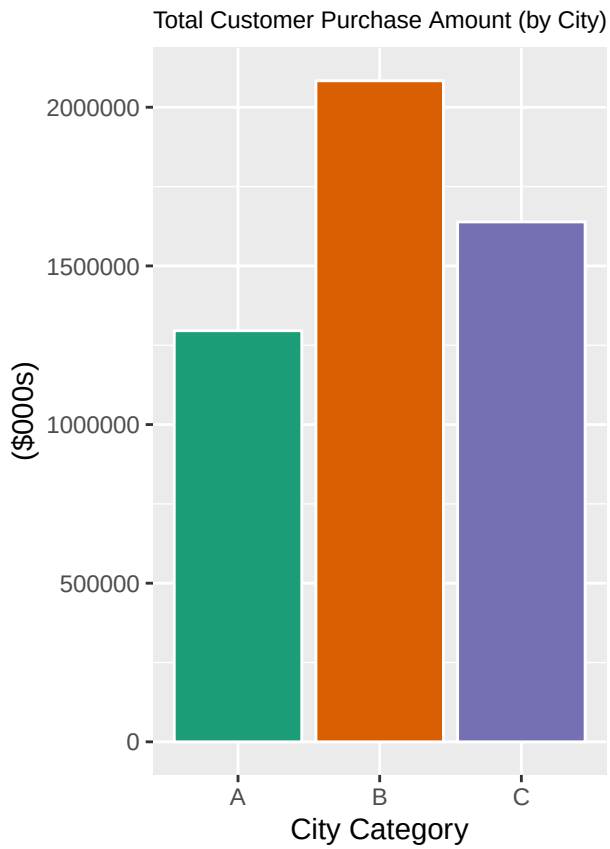
```
## # A tibble: 6 x 2
## # Groups:   User_ID [6]
##   User_ID     n
##     <int> <int>
## 1 1000001    34
## 2 1000002    76
## 3 1000003    29
## 4 1000004    13
## 5 1000005   106
## 6 1000006    46
```

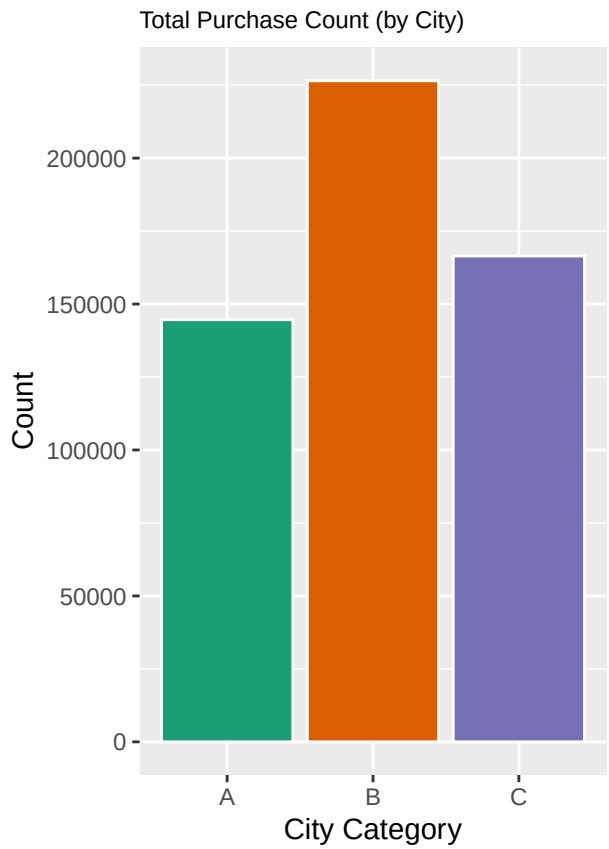
This calculation gives us the total number of purchases made by each individual user. To further investigate this, we should also calculate the total purchase amount made by each user. Once we have this information, we can join it with the other table to gain a more complete understanding of our findings.

```
## # A tibble: 6 x 3
##   User_ID City_Category     n
##     <int> <chr>         <int>
## 1 1000001 A                34
```

```
## 2 1000002 C                76
## 3 1000003 A                29
## 4 1000004 B                13
## 5 1000005 A               106
## 6 1000006 A                46
```

```
## # A tibble: 3 x 2
##   City_Category CountOfPurchases
##   <chr>                    <int>
## 1 A                       144638
## 2 B                       226493
## 3 C                       166446
```





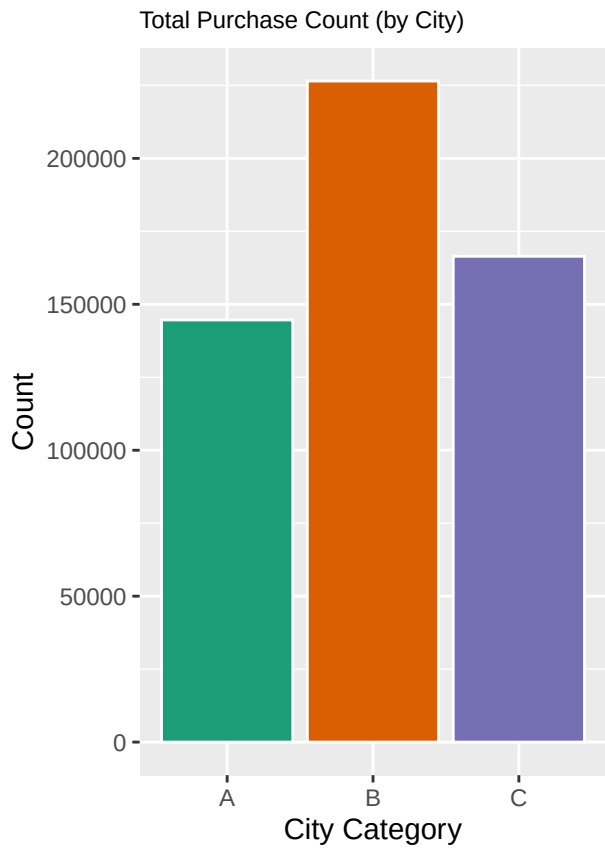
Based on these charts, it seems that customers from City B are not necessarily buying more expensive products than those from City A and City C. Instead, they are simply making more purchases in general. This inference is supported by the fact that the "Total Count of Purchases" chart has a similar shape to the "Total Customer Purchase Amount" chart. If the opposite were true, we would expect to see a lower count of total purchases for customers from City B, accompanied by a higher total purchase amount.

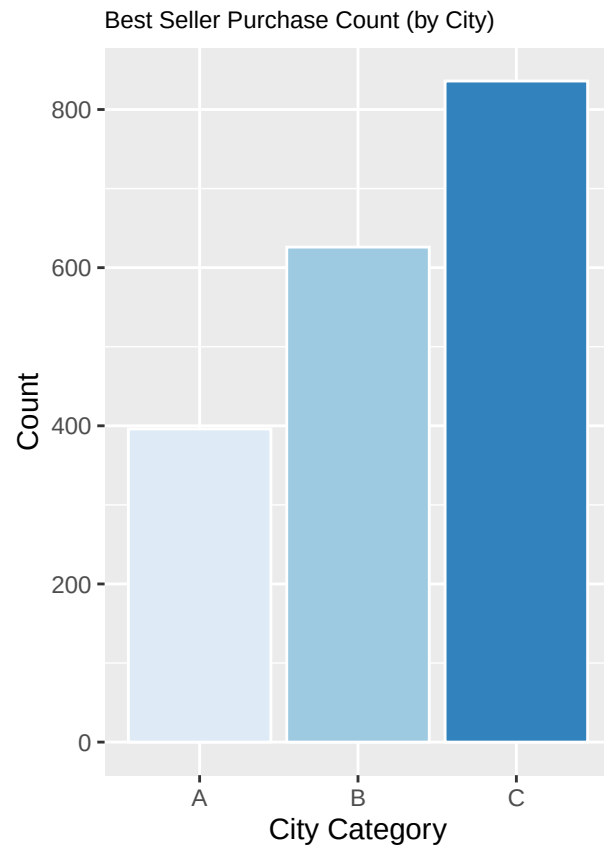
Given this observation, let's now turn our attention to the distribution of our best-selling product (P00265242) within each City_Category.

```
##      User_ID Product_ID Gender   Age Occupation City_Category
## 400  1000066  P00265242      M 26-35         18             C
## 1192 1000196  P00265242      F 36-45          9             C
## 1373 1000222  P00265242      M 26-35          1             A
## 1846 1000301  P00265242      M 18-25          4             B
## 2210 1000345  P00265242      M 26-35         12             A
## 2405 1000383  P00265242      F 26-35          7             A
##      Stay_In_Current_City_Years Marital_Status Product_Category_1
## 400                           2              0                  5
## 1192                         4+              0                  5
## 1373                          1              0                  5
## 1846                         4+              0                  5
## 2210                          2              1                  5
## 2405                         4+              1                  5
##      Product_Category_2 Product_Category_3 Purchase
## 400                   8                 NA     8652
## 1192                  8                 NA     8767
## 1373                  8                 NA     6944
```

```
## 1846                    8                  NA      8628
## 2210                    8                  NA      8593
## 2405                    8                  NA      6998
```

```
##   City_Category   n
## 1             A 396
## 2             B 626
## 3             C 836
```





While customers in City C purchase more of our "best seller" product than those in City A and City B, the overall number of purchases made by City C residents is lower than that of City B.

2.5 Stay in Current City

Next, we will investigate the distribution of customers based on the length of time they have lived in their respective cities.

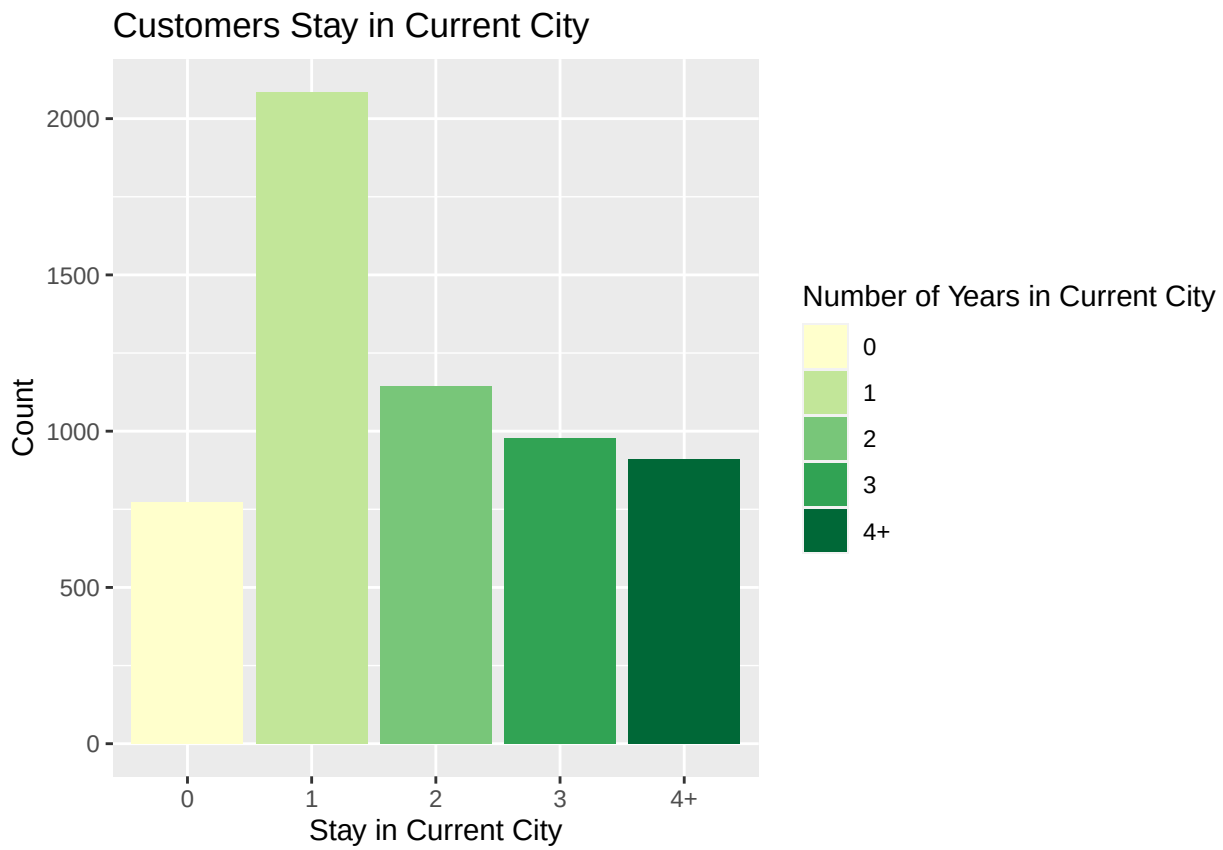
```
## # A tibble: 6 x 3
## # Groups:   User_ID [6]
##   User_ID City_Category Stay_In_Current_City_Years
##     <int> <chr>         <chr>
## 1 1000001 A             2
## 2 1000002 C             4+
## 3 1000003 A             3
## 4 1000004 B             2
```

```
## 5 1000005 A             1
## 6 1000006 A             1
```

let's begin by plotting the distribution of our customers across the different cities.

```
## # A tibble: 3 x 2
##   City_Category     n
##   <chr>         <int>
## 1 A              1045
## 2 B              1707
## 3 C              3139
```

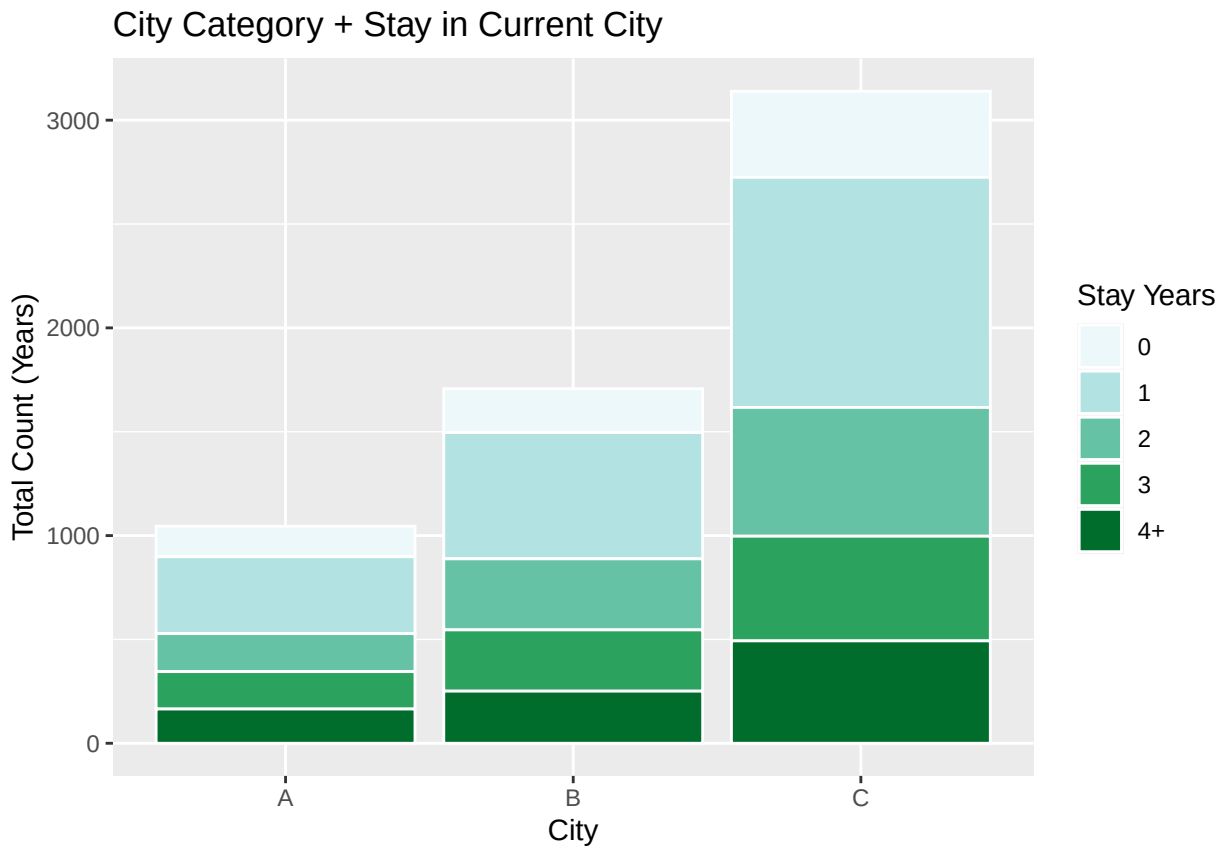
Based on the chart, it appears that a large proportion of our customer base resides in City C. Let's dive deeper into this trend and explore it further.



let's create a stacked bar chart to visualize the distribution of customers based on their length of residence in their respective cities, segmented by City_Category.

```
## # A tibble: 6 x 4
## # Groups:   City_Category [2]
##   City_Category Stay_In_Current_City_Years     n Percentage
##   <chr>         <chr>                      <int>      <dbl>
## 1 A             0                            147       14.1
## 2 A             1                            370       35.4
## 3 A             2                            183       17.5
## 4 A             3                            180       17.2
```

```
## 5 A             4+                          165       15.8
## 6 B             0                           211       12.4
```



Based on the stacked bar chart, it appears that the majority of our customers across all City_Categories have only been living in their respective cities for 1 year. It would be interesting to further investigate the reasons for this trend and see if there are any patterns or factors contributing to it.

2.6 Purchase

Let's calculate the total purchase amount for each user in the dataset. This will give us an idea of the most valuable customers for our store.

```
## # A tibble: 6 x 2
##   User_ID Purchase_Amount
##     <int>           <int>
## 1 1000001          333481
## 2 1000002          810353
## 3 1000003          341635
## 4 1000004          205987
## 5 1000005          821001
## 6 1000006          379450
```

now we can sort the total purchase amounts by user ID in descending order to find our top spenders. This will help us identify the customers who make the most purchases and contribute the most to our store's revenue.

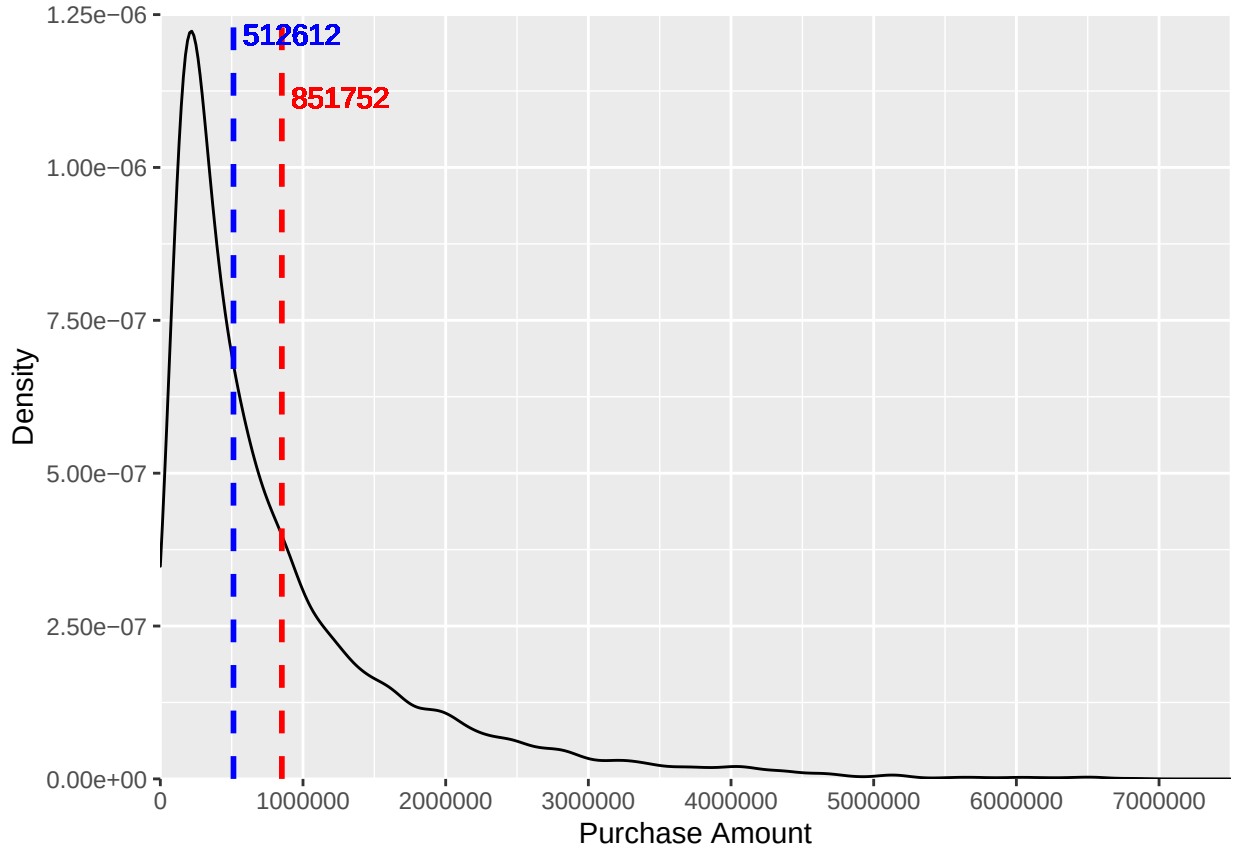
```
## # A tibble: 6 x 2
##   User_ID Purchase_Amount
##     <int>           <int>
## 1 1004277        10536783
## 2 1001680         8699232
## 3 1002909         7577505
## 4 1001941         6817493
## 5 1000424         6573609
## 6 1004448         6565878
```

Based on our analysis, we have identified User ID 1004277 as the top spender at our store. In order to gain more insights into our customer spending data, we will use the summary() function to see other important characteristics such as the mean, median, and quartiles of our total purchase amount data. This will give us a better understanding of the typical spending behavior of our customers, as well as any outliers or extreme values in our data set.

```
##     User_ID        Purchase_Amount
##  Min.   :1000001   Min.   :   44108
##  1st Qu.:1001518   1st Qu.:  234914
##  Median :1003026   Median :  512612
##  Mean   :1003025   Mean   :  851752
##  3rd Qu.:1004532   3rd Qu.: 1099005
##  Max.   :1006040   Max.   :10536783
```

We have computed the total purchase amount by User ID and found that User ID 1004277 is our top spender. By using the summary() function on the total customer spending data, we can see that the average total purchase amount is 851752, with a maximum total purchase amount of 10536783 and a minimum total purchase amount of 44108. The median purchase amount is 512612.

To better understand the distribution of purchase amounts, we will plot a density chart. This will allow us to see where the highest concentration of similar purchase amounts lies within the entire customer base. It is important to note that density charts represent the expected probability of values given the input data, and plot a line surrounding those values to estimate the distribution.



This density plot of purchase amounts shows that the distribution is right-skewed, with a long tail towards the right. This indicates that there are a significant number of purchases with higher values than the mean, and that the distribution of purchases is not normally distributed. The highest density of values appears to be around the 250000 mark, with a steep drop-off in density for values above this point. Overall, this suggests that there are a significant number of high-value purchases made by our customers, which may be worth investigating further.

2.7 Marital Status

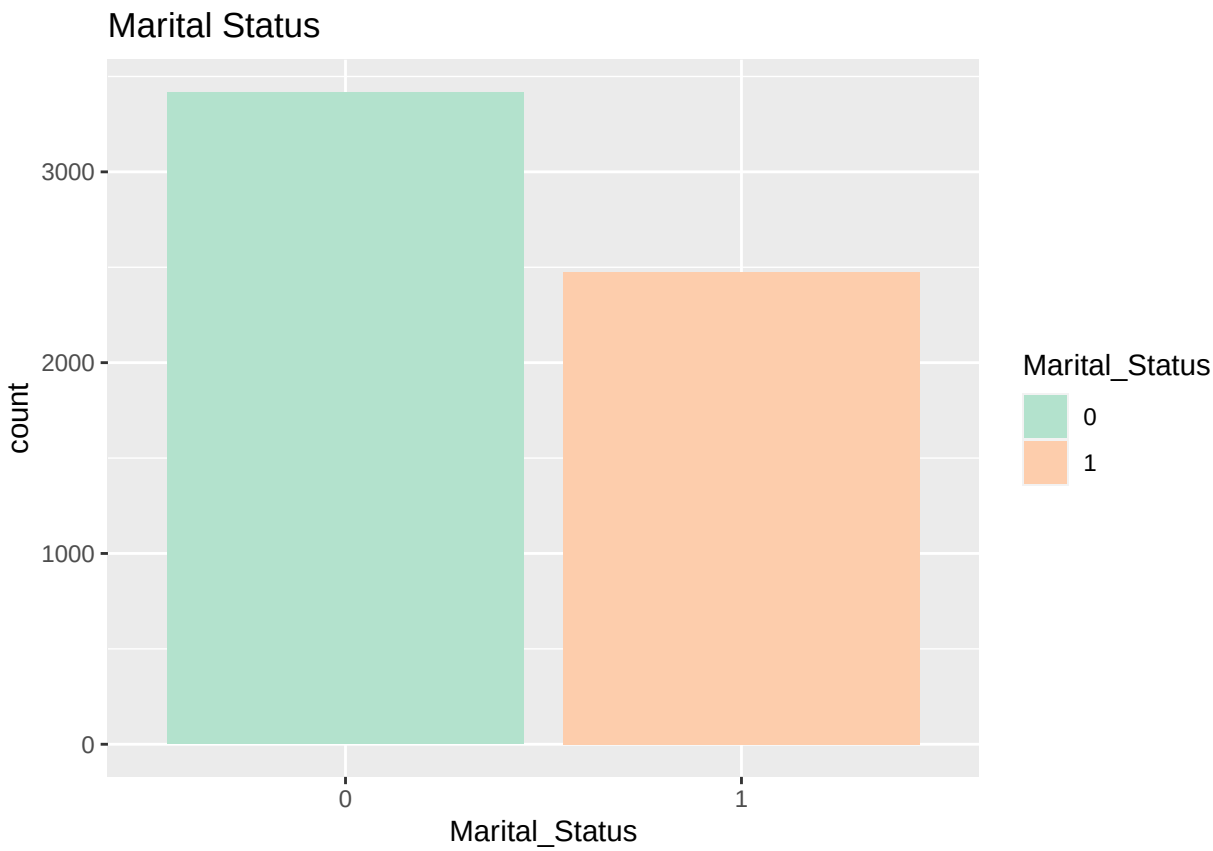
Lets now examine the marital status of store customers.

```
## # A tibble: 6 x 2
## # Groups:   User_ID [6]
##   User_ID Marital_Status
##     <int>          <int>
## 1 1000001              0
## 2 1000002              0
## 3 1000003              0
## 4 1000004              1
## 5 1000005              1
## 6 1000006              0
```

Note, we need to quickly change Marital_Status from a numeric variable to a categorical type.

```
## [1] "character"
```

It's important to note that we don't have a clear guide for the marital status variable in the dataset. While in some cases it would be best to contact the data provider for clarification, in this case we can make the assumption that the values of 1 represent "married" and the values of 0 represent "single". To make further analysis easier, we will convert the variable from numeric to categorical.

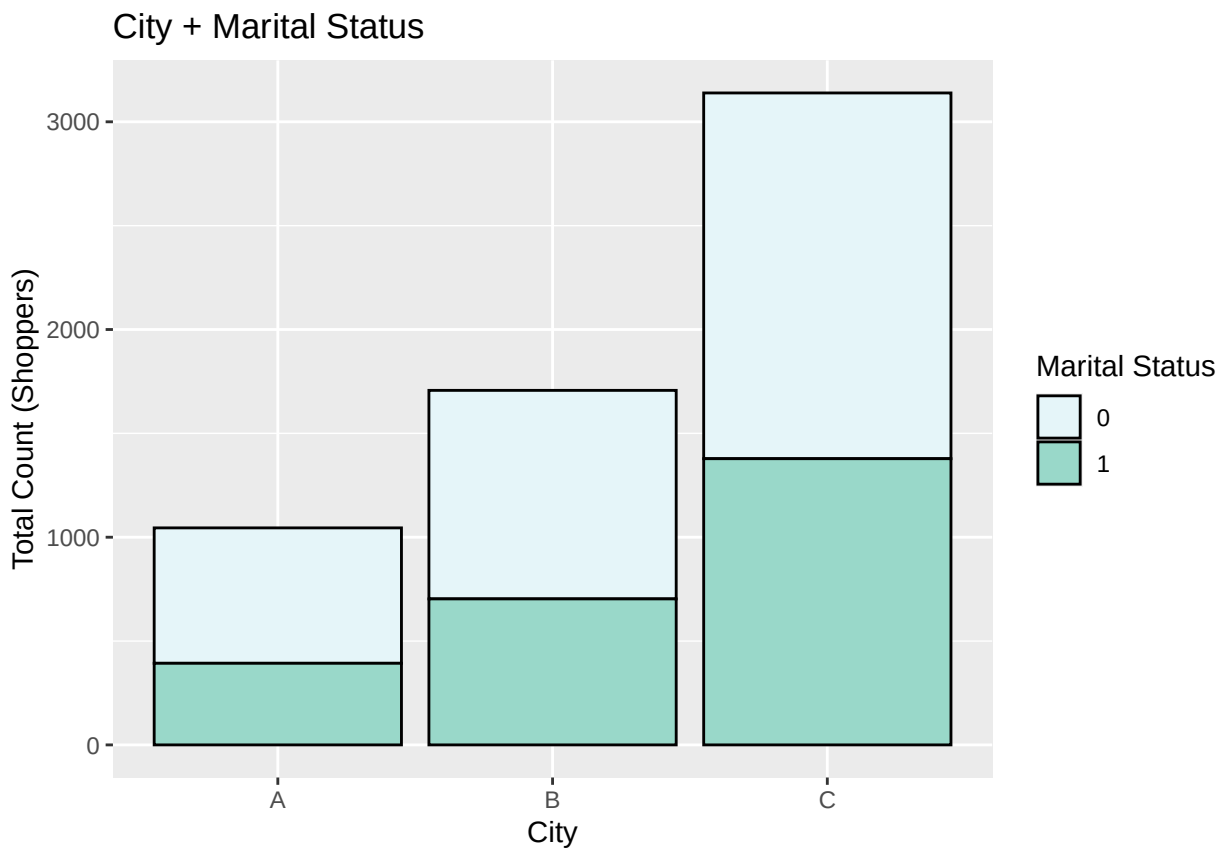


Let's create a stacked bar chart of Marital_Status in each City_Category to see if there are any differences.

```
## # A tibble: 6 x 4
## # Groups:   User_ID [6]
##   User_ID Marital_Status City_Category Stay_In_Current_City_Years
##     <int> <chr>          <chr>         <chr>
## 1 1000001 0              A             2
## 2 1000002 0              C             4+
## 3 1000003 0              A             3
## 4 1000004 1              B             2
## 5 1000005 1              A             1
## 6 1000006 0              A             1
```

```
## # A tibble: 6 x 3
## # Groups:   City_Category [3]
##   City_Category Marital_Status     n
##   <chr>         <chr>          <int>
## 1 A             0                652
## 2 A             1                393
## 3 B             0               1004
## 4 B             1                703
```

```
## 5 C             0                1761
## 6 C             1                1378
```



let's investigate the Stay_in_Current_City distribution within each City_Category.

```
##   User_ID   Age
## 1 1000001  0-17
## 2 1000002   55+
## 3 1000003 26-35
## 4 1000004 46-50
## 5 1000005 26-35
## 6 1000006 51-55
```

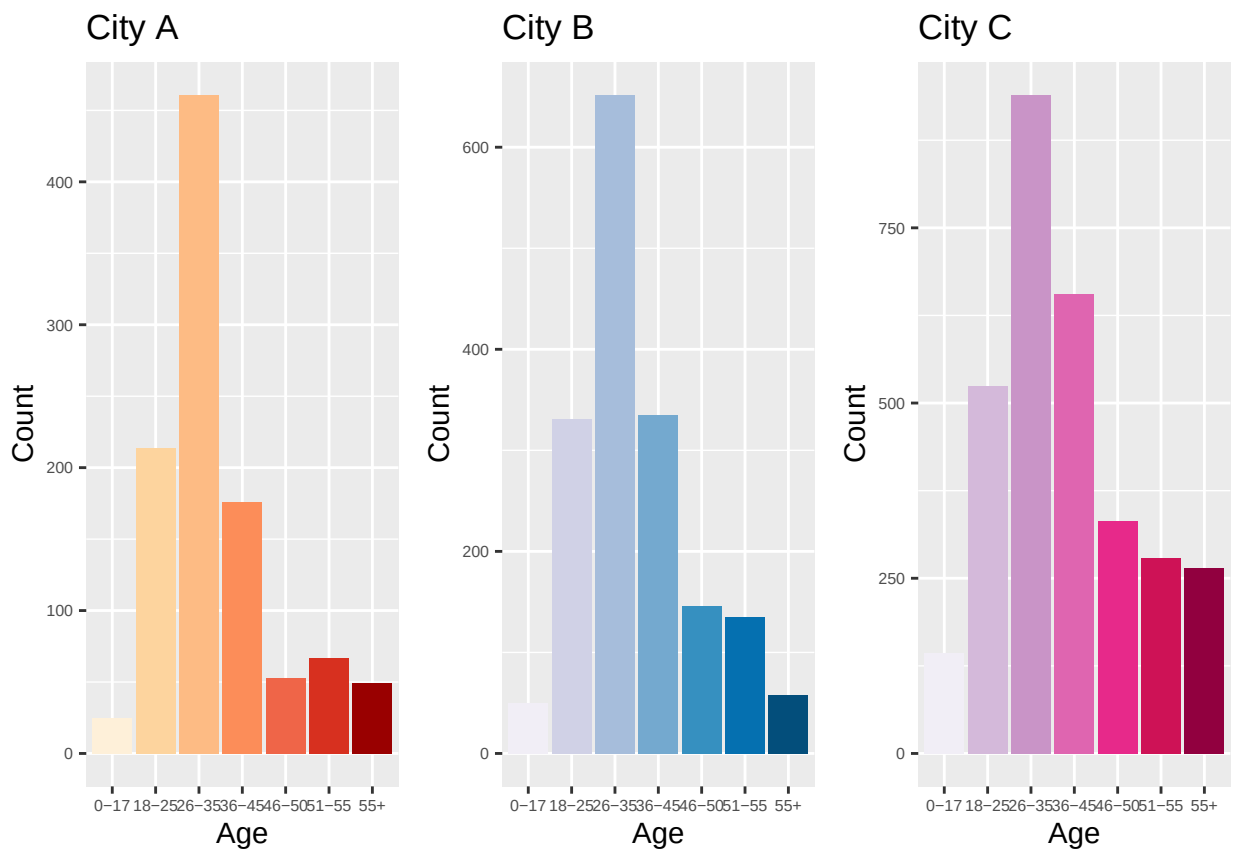
```
## # A tibble: 6 x 5
## # Groups:   User_ID [6]
##   User_ID Marital_Status City_Category Stay_In_Current_City_Years Age
##     <int> <chr>          <chr>         <chr>                      <chr>
## 1 1000001 0              A             2                          0-17
## 2 1000002 0              C             4+                         55+
## 3 1000003 0              A             3                          26-35
## 4 1000004 1              B             2                          46-50
## 5 1000005 1              A             1                          26-35
## 6 1000006 0              A             1                          51-55
```

```
## # A tibble: 6 x 5
```

```
## # Groups:   User_ID [6]
##   User_ID Marital_Status City_Category Stay_In_Current_City_Years Age
##     <int> <chr>          <chr>         <chr>                       <chr>
## 1 1000001 0              A             2                           0-17
## 2 1000003 0              A             3                           26-35
## 3 1000005 1              A             1                           26-35
## 4 1000006 0              A             1                           51-55
## 5 1000015 0              A             1                           26-35
## 6 1000019 0              A             3                           0-17
```

```
## # A tibble: 6 x 5
## # Groups:   User_ID [6]
##   User_ID Marital_Status City_Category Stay_In_Current_City_Years Age
##     <int> <chr>          <chr>         <chr>                       <chr>
## 1 1000004 1              B             2                           46-50
## 2 1000007 1              B             1                           36-45
## 3 1000010 1              B             4+                          36-45
## 4 1000018 0              B             3                           18-25
## 5 1000021 0              B             0                           18-25
## 6 1000023 1              B             3                           36-45
```

```
## # A tibble: 6 x 5
## # Groups:   User_ID [6]
##   User_ID Marital_Status City_Category Stay_In_Current_City_Years Age
##     <int> <chr>          <chr>         <chr>                       <chr>
## 1 1000002 0              C             4+                          55+
## 2 1000008 1              C             4+                          26-35
## 3 1000009 0              C             0                           26-35
## 4 1000011 0              C             1                           26-35
## 5 1000012 0              C             2                           26-35
## 6 1000013 1              C             3                           46-50
```



Another factor that could contribute to the difference in Marital_Status between cities could be cultural and socioeconomic differences. It would be interesting to further investigate these factors to understand the underlying reasons for the observed patterns.

2.8 Top Shoppers

Now, we will investigate who our top shoppers were on Black Friday.

```
##   User_ID    n
## 1 1001680 1025
## 2 1004277  978
## 3 1001941  898
## 4 1001181  861
## 5 1000889  822
## 6 1003618  766
```

According to our analysis, User_ID 1001680 appears to be the most frequent shopper in our master ledger of shopper data. It seems that this user has made over 1000 individual transactions based on the data. To gain more insights, we can combine this top shoppers dataset with our total customer purchases dataset, allowing us to see the shopping behavior and purchase patterns of the top spenders in more detail.

```
##   User_ID    n Purchase_Amount
## 1 1001680 1025         8699232
## 2 1004277  978        10536783
## 3 1001941  898         6817493
```

```
## 4 1001181  861         6387899
## 5 1000889  822         5499812
## 6 1003618  766         5961987
```

After merging our top shoppers dataset with the total customer purchases dataset, we can observe that User_ID 1001680 has made the most number of purchases, with over 1000 transactions recorded. However, our previous analysis showed that User_ID 1004277 has the highest total Purchase_Amount. Moreover, we can also compute the average Purchase_Amount for each user using this merged dataset.

```
##   User_ID    n Purchase_Amount Average_Purchase_Amount
## 1 1001680 1025         8699232                8487.056
## 2 1004277  978        10536783               10773.807
## 3 1001941  898         6817493                7591.863
## 4 1001181  861         6387899                7419.163
## 5 1000889  822         5499812                6690.769
## 6 1003618  766         5961987                7783.273
```

Let's sort the data based on the Average_Purchase_Amount column to identify which customers are spending the most on average. This will allow us to gain insights into which customers are the most valuable to the business in terms of their spending habits.

```
##   User_ID  n Purchase_Amount Average_Purchase_Amount
## 1 1005069 16          308454                19278.38
## 2 1003902 93         1746284                18777.25
## 3 1005999 18          330227                18345.94
## 4 1001349 23          417743                18162.74
## 5 1000101 65         1138239                17511.37
## 6 1003461 20          350174                17508.70
```

It seems that User_ID 1005069 is the customer with the highest average purchase amount, with a total purchase amount of 308454. On the other hand, User_ID 1003902 is the second-highest in average purchase amount, but with a much higher total purchase amount of 1746284.

2.9 Occupation

The last thing we will analyze is the occupation of customers in our dataset.

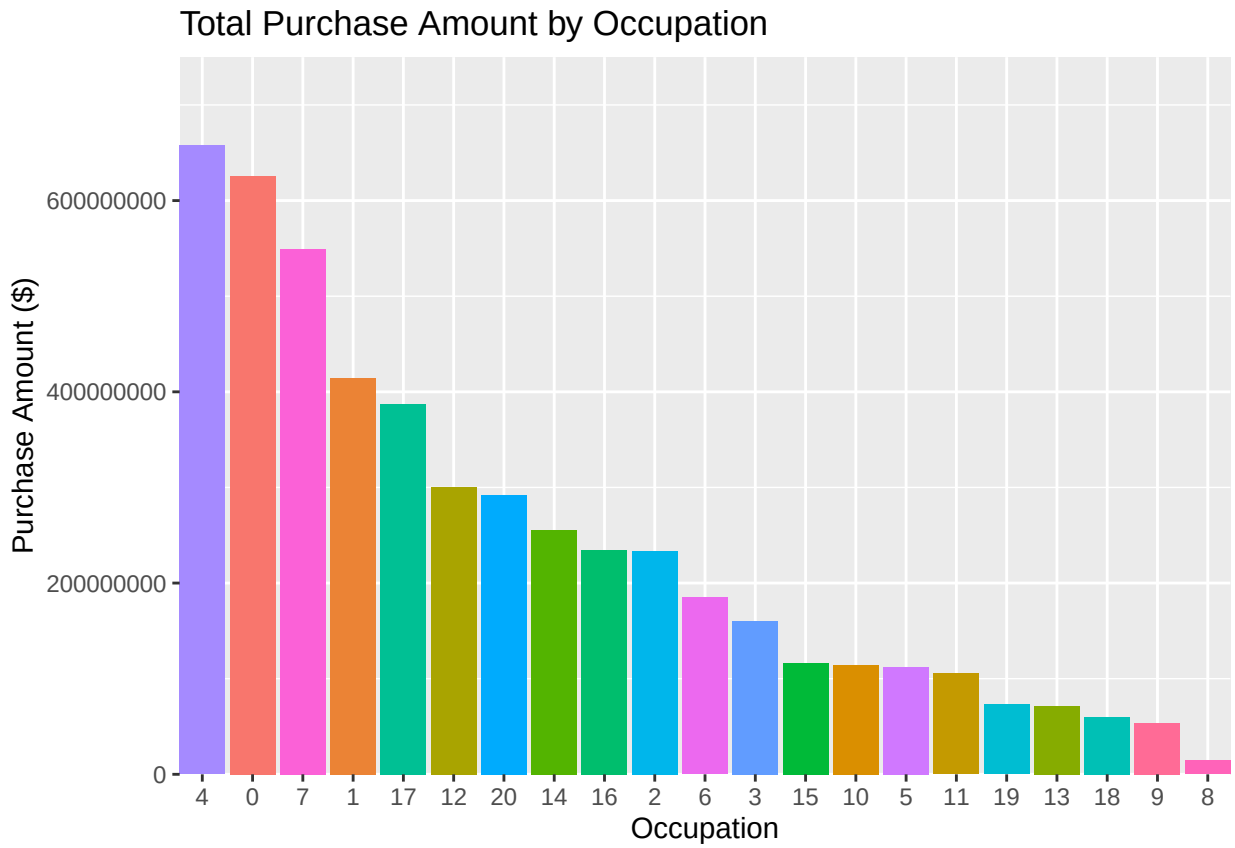
```
## # A tibble: 6 x 3
## # Groups:   User_ID [6]
##   User_ID Occupation Purchase_Amount
##     <int>      <int>           <int>
## 1 1000001         10          333481
## 2 1000002         16          810353
## 3 1000003         15          341635
## 4 1000004          7          205987
## 5 1000005         20          821001
## 6 1000006          9          379450
```

Now that we have our required dataset, we can aggregate the total Purchase_Amount for each Occupation identifier. Then, we will convert the Occupation column to a character data type.

```
## [1] "character"
```

```
## # A tibble: 6 x 2
##   Occupation Purchase_Amount
##   <chr>                <int>
## 1 4                657530393
## 2 0                625814811
## 3 7                549282744
## 4 1                414552829
## 5 17               387240355
## 6 12               300672105
```

Now, lets plot each occupation and their total Purchase_Amount



By grouping total purchase amount by occupation, we can see that customers with Occupation 4 spent the most during Black Friday. Customers with Occupation 0 and 7 also spent significantly high amounts. If we had access to the key for the occupations, we could use this information to classify our shoppers and create targeted marketing campaigns based on their spending behavior.

3 Methods and Analysis

3.1 Apriori (Association Rule Learning)

Let's explore the use of the Apriori algorithm, which is a machine learning technique used to make association rules related to customer purchases. We will be utilizing the arules package to perform this analysis.

The fundamental concept behind Association Rule Learning is to predict customer transactions based on their prior purchases. It aims to answer the question "People who bought —— also bought ——". This

approach can be immensely valuable for retailers who want to optimize their product placement strategy in stores and promotional campaigns.

In the context of our Black Friday store, implementing an effective product placement strategy can maximize the sales of related products that are often purchased together. For instance, if our store is offering a discount on TVs, it would be wise to place HDMI cables next to them, as these items are frequently purchased together. However, placing these items far apart may also be beneficial as it would require customers to navigate the entire store, increasing the likelihood of them noticing and purchasing other products along the way.

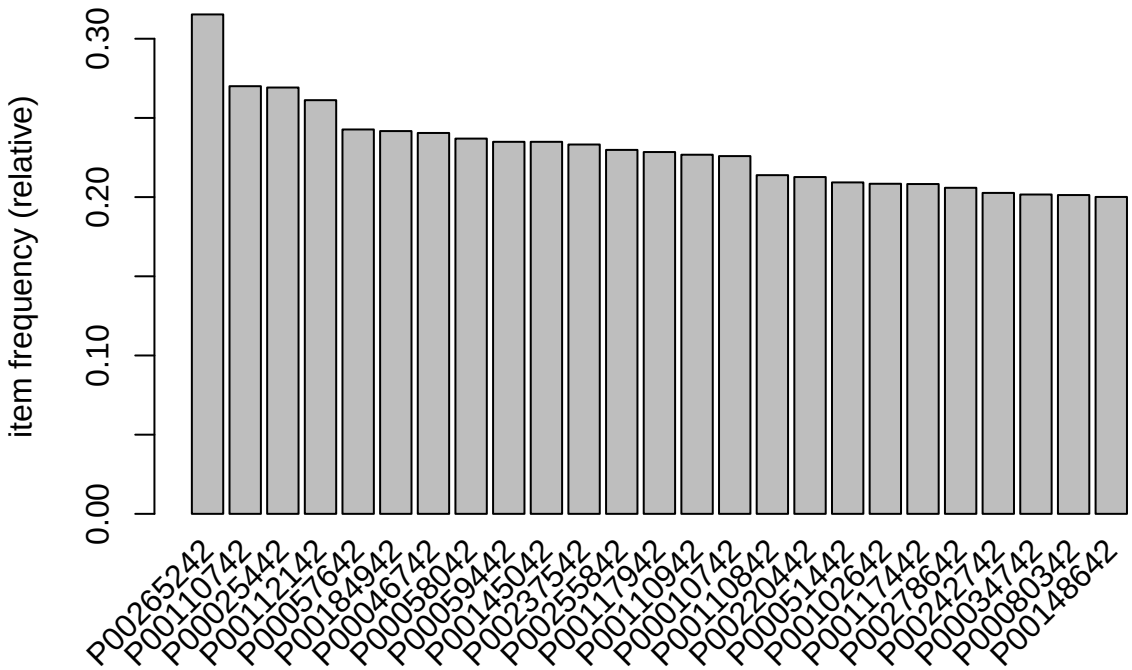
The Apriori algorithm is particularly adept at maximizing the likelihood of customers performing, purchasing, or watching something based on their prior actions.

```
## distribution of transactions with duplicates:
## items
##   46  126  163  202  258  272  285  307  310  316  319  327  330  334  340  344
##    1    1    1    1    1    1    1    1    1    1    1    2    1    1    1    1
##  345  348  354  357  373  393  402  408  419  437  441  449  450  452  454  456
##    1    1    1    1    1    1    1    1    1    1    1    1    1    1    1    1
##  459  465  466  467  475  476  477  481  487  491  495  498  507  523  524  526
##    1    1    2    2    1    1    1    1    2    2    3    2    1    1    2    1
##  527  528  530  531  532  533  535  537  538  539  540  545  546  548  549  553
##    2    1    1    2    1    1    1    1    2    3    1    1    1    3    1    1
##  554  555  556  558  563  566  567  570  572  574  575  577  578  580  583  584
##    2    1    1    2    1    3    1    3    2    4    1    2    3    2    1    1
##  586  588  589  590  591  592  593  594  595  597  598  601  602  604  607  608
##    2    3    5    1    1    1    1    1    3    2    1    1    1    1    2    1
##  610  612  613  614  615  616  617  618  619  620  623  625  632  633  634  635
##    1    2    1    2    1    1    2    1    3    2    1    1    6    1    5    2
##  638  640  641  642  643  644  645  646  647  648  653  654  657  658  659  661
##    2    2    1    2    1    2    3    1    4    1    1    3    2    1    1    2
##  662  663  664  665  666  667  668  669  670  671  672  674  676  677  678  679
##    1    3    1    1    2    1    2    4    2    1    3    1    1    2    3    3
##  681  682  683  685  686  687  688  689  690  691  692  694  695  697  698  699
##    2    2    4    4    5    2    2    1    1    1    1    2    1    1    1    1
##  700  702  703  704  705  706  707  708  709  710  712  713  714  715  716  717
##    2    1    3    1    1    2    2    3    2    2    4    5    2    2    1    1
##  718  719  720  721  722  723  724  725  726  727  728  729  730  732  733  734
##    2    3    3    5    2    1    1    2    5    2    1    3    3    2    1    5
##  735  736  737  738  739  740  741  742  743  744  745  747  748  749  750  751
##    6    2    4    6    3    4    1    6    7    4    6    1    1    5    5    3
##  752  753  754  755  756  757  758  759  760  761  763  764  765  766  767  768
##    2    3    4    6    6    2    6    2    1    5    3    4    2    3    2    5
##  769  770  771  772  773  774  775  776  777  778  779  780  781  782  783  784
##    2    2    3    1    3    4    2    2    3    3    2    4    7    3    4    5
##  785  786  787  788  789  790  791  792  793  794  795  796  797  798  799  800
##    5    3    3    6    5    5    2    3    5    3    8    5    5    9    3    4
##  801  802  803  804  805  806  807  808  809  810  811  812  813  814  815  816
##    4    7    4    3    4    5    7    5    4    5    3    2    6    6    3   11
##  817  818  819  820  821  822  823  824  825  826  827  828  829  830  831  832
##    5   10    6    6    4    7    7    2    5    4    7    5    5    4    5    4
##  833  834  835  836  837  838  839  840  841  842  843  844  845  846  847  848
##    3    5    4   11    5    5    4    9    7    6    4    7    9   11    4    6
##  849  850  851  852  853  854  855  856  857  858  859  860  861  862  863  864
##   10    6   10    7   12   16   11    8    7    4   12    9   11   11    9    6
##  865  866  867  868  869  870  871  872  873  874  875  876  877  878  879  880
```

```
##   11   10    7    6    5   12    6    7    8   11    9    9    8    7    5    4
##  881  882  883  884  885  886  887  888  889  890  891  892  893  894  895  896
##   15   13   12    8    4    6   12   15   13   10   11   13    6   21    7   14
##  897  898  899  900  901  902  903  904  905  906  907  908  909  910  911  912
##    9    7   11   18    5   14   10    9   19   15   10   17   18   23    8   19
##  913  914  915  916  917  918  919  920  921  922  923  924  925  926  927  928
##   15   12   18   21   17   12   11   13   13   12   20   20   16   13   15   17
##  929  930  931  932  933  934  935  936  937  938  939  940  941  942  943  944
##   27   22   20   28   18   14   20   20   20   14   22   30   23   23   21   20
##  945  946  947  948  949  950  951  952  953  954  955  956  957  958  959  960
##   25   19   30   31   30   24   27   25   40   30   31   16   29   30   32   48
##  961  962  963  964  965  966  967  968  969  970  971  972  973  974  975  976
##   27   27   24   30   26   35   43   30   51   49   40   41   36   32   36   38
##  977  978  979  980  981  982  983  984  985  986  987  988  989  990  991  992
##   43   41   42   37   49   44   51   57   55   40   53   56   63   39   58   50
##  993  994  995  996  997  998  999 1000 1001 1002 1003 1004 1005 1006 1007 1008
##   58   77   74   72   72   84   74   66   77   85   93   79   94  118  122  104
## 1009 1010 1011 1012 1013 1014 1015 1016 1017 1018 1019
##  121  113  120   78   77   55   37   20    7    5    1
```

```
## transactions as itemMatrix in sparse format with
##  5892 rows (elements/itemsets/transactions) and
##  10539 columns (items) and a density of 0.008768598
##
## most frequent items:
## P00265242 P00110742 P00025442 P00112142 P00057642   (Other)
##      1858      1591      1586      1539      1430    536489
##
## element (itemset/transaction) length distribution:
## sizes
##    6    7    8    9   10   11   12   13   14   15   16   17   18   19   20   21
##    1    5    7   20   37   55   77   78  120  113  121  104  122  118   94   79
##   22   23   24   25   26   27   28   29   30   31   32   33   34   35   36   37
##   93   85   77   66   74   84   72   72   74   77   58   50   58   39   63   56
##   38   39   40   41   42   43   44   45   46   47   48   49   50   51   52   53
##   53   40   55   57   51   44   49   37   42   41   43   38   36   32   36   41
##   54   55   56   57   58   59   60   61   62   63   64   65   66   67   68   69
##   40   49   51   30   43   35   26   30   24   27   27   48   32   30   29   16
##   70   71   72   73   74   75   76   77   78   79   80   81   82   83   84   85
##   31   30   40   25   27   24   30   31   30   19   25   20   21   23   23   30
##   86   87   88   89   90   91   92   93   94   95   96   97   98   99  100  101
##   22   14   20   20   20   14   18   28   20   22   27   17   15   13   16   20
##  102  103  104  105  106  107  108  109  110  111  112  113  114  115  116  117
##   20   12   13   13   11   12   17   21   18   12   15   19    8   23   18   17
##  118  119  120  121  122  123  124  125  126  127  128  129  130  131  132  133
##   10   15   19    9   10   14    5   18   11    7    9   14    7   21    6   13
##  134  135  136  137  138  139  140  141  142  143  144  145  146  147  148  149
##   11   10   13   15   12    6    4    8   12   13   15    4    5    7    8    9
##  150  151  152  153  154  155  156  157  158  159  160  161  162  163  164  165
##    9   11    8    7    6   12    5    6    7   10   11    6    9   11   11    9
##  166  167  168  169  170  171  172  173  174  175  176  177  178  179  180  181
##   12    4    7    8   11   16   12    7   10    6   10    6    4   11    9    7
##  182  183  184  185  186  187  188  189  190  191  192  193  194  195  196  197
##    4    6    7    9    4    5    5   11    4    5    3    4    5    4    5    5
```

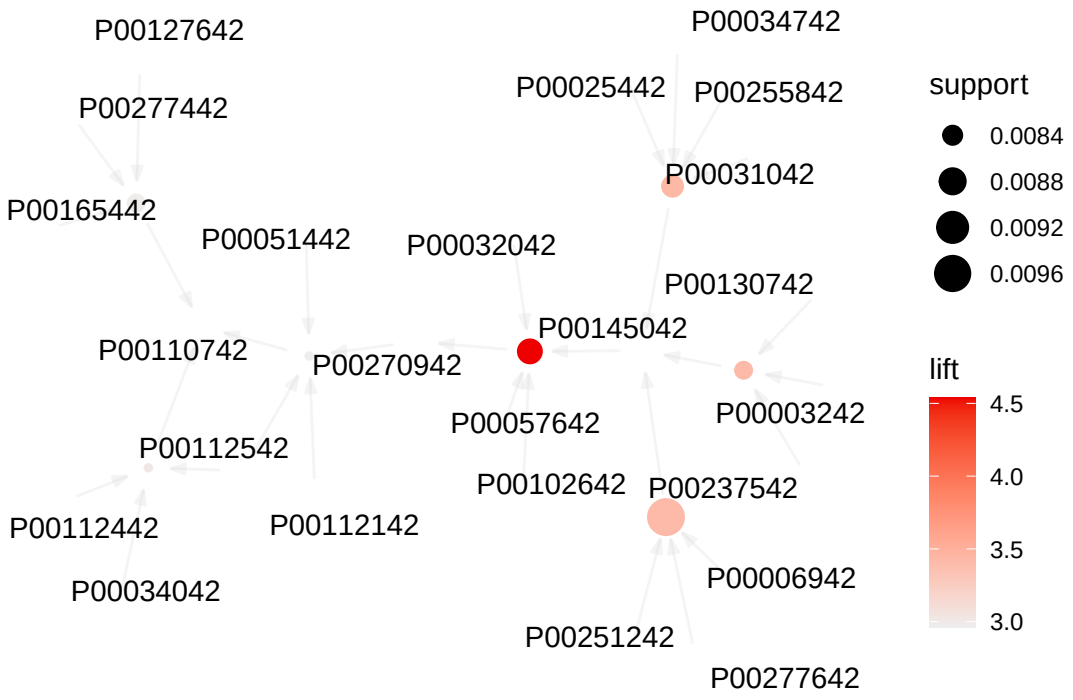
```
##  198  199  200  201  202  203  204  205  206  207  208  209  210  211  212  213
##    7    4    5    2    7    7    4    6    6   10    5   11    3    6    6    2
##  214  215  216  217  218  219  220  221  222  223  224  225  226  227  228  229
##    3    5    4    5    7    5    4    3    4    7    4    4    3    9    5    5
##  230  231  232  233  234  235  236  237  238  239  240  241  242  243  244  245
##    8    3    5    3    2    5    5    6    3    3    5    5    4    3    7    4
##  246  247  248  249  250  251  252  253  254  255  256  257  258  259  260  261
##    2    3    3    2    2    4    3    1    3    2    2    5    2    3    2    4
##  262  264  265  266  267  268  269  270  271  272  273  274  275  276  277  278
##    3    5    1    2    6    2    6    6    4    3    2    3    5    5    1    1
##  280  281  282  283  284  285  286  287  288  289  290  291  292  293  295  296
##    6    4    7    6    1    4    3    6    4    2    6    5    1    2    3    3
##  297  298  299  300  301  302  303  304  305  306  307  308  309  310  311  312
##    1    2    5    2    1    1    2    5    3    3    2    1    1    2    2    5
##  313  315  316  317  318  319  320  321  322  323  325  326  327  328  330  331
##    4    2    2    3    2    2    1    1    3    1    2    1    1    1    1    2
##  333  334  335  336  337  338  339  340  342  343  344  346  347  348  349  351
##    1    1    1    1    2    2    5    4    4    2    2    3    3    2    1    1
##  353  354  355  356  357  358  359  360  361  362  363  364  366  367  368  371
##    3    1    2    4    2    1    2    1    1    3    1    2    1    1    2    3
##  372  377  378  379  380  381  382  383  384  385  387  390  391  392  393  400
##    1    1    4    1    3    2    1    2    1    2    2    2    5    1    6    1
##  402  405  406  407  408  409  410  411  412  413  415  417  418  421  423  424
##    1    2    3    1    2    1    1    2    1    2    1    1    2    1    1    1
##  427  428  430  431  432  433  434  435  436  437  439  441  442  445  447  448
##    1    2    3    1    1    1    1    1    5    3    2    1    1    2    3    2
##  450  451  453  455  458  459  462  467  469  470  471  472  476  477  479  480
##    1    4    2    3    1    3    1    2    1    1    2    1    1    3    1    1
##  485  486  487  488  490  492  493  494  495  497  498  499  501  502  518  527
##    1    3    2    1    1    1    1    2    1    1    2    1    2    1    1    2
##  530  534  538  544  548  549  550  558  559  560  566  569  571  573  575  576
##    3    2    2    1    1    1    1    2    2    1    1    1    1    1    1    1
##  584  588  606  617  623  632  652  668  671  677  680  681  685  691  695  698
##    1    1    1    1    1    1    1    1    1    1    1    1    1    1    1    2
##  706  709  715  718  740  753  767  823  862  899  979 1025 1026
##    1    1    1    1    1    1    1    1    1    1    1    1    1
##
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    6.00   26.00   54.00   92.41  115.00 1026.00
##
## includes extended item information - examples:
##    labels
## 1 1000001
## 2 1000002
## 3 1000003
```




```
## Apriori
##
## Parameter specification:
##  confidence minval smax arem  aval originalSupport maxtime support minlen
##         0.8    0.1    1 none FALSE            TRUE       0   0.008      1
##  maxlen target  ext
##      10  rules TRUE
##
## Algorithmic control:
##  filter tree heap memopt load sort verbose
##     0.1 TRUE TRUE  FALSE TRUE    2    TRUE
##
## Absolute minimum support count: 47
##
## set item appearances ...[0 item(s)] done [0.00s].
## set transactions ...[10539 item(s), 5892 transaction(s)] done [0.55s].
## sorting and recoding items ... [2099 item(s)] done [0.02s].
## creating transaction tree ... done [0.01s].
## checking subsets of size 1 2 3 4 5 6 done [55.47s].
## writing ... [7 rule(s)] done [1.17s].
## creating S4 object  ... done [1.07s].

##     lhs             rhs             support confidence    coverage     lift count
## [1] {P00032042,
##      P00057642,
```

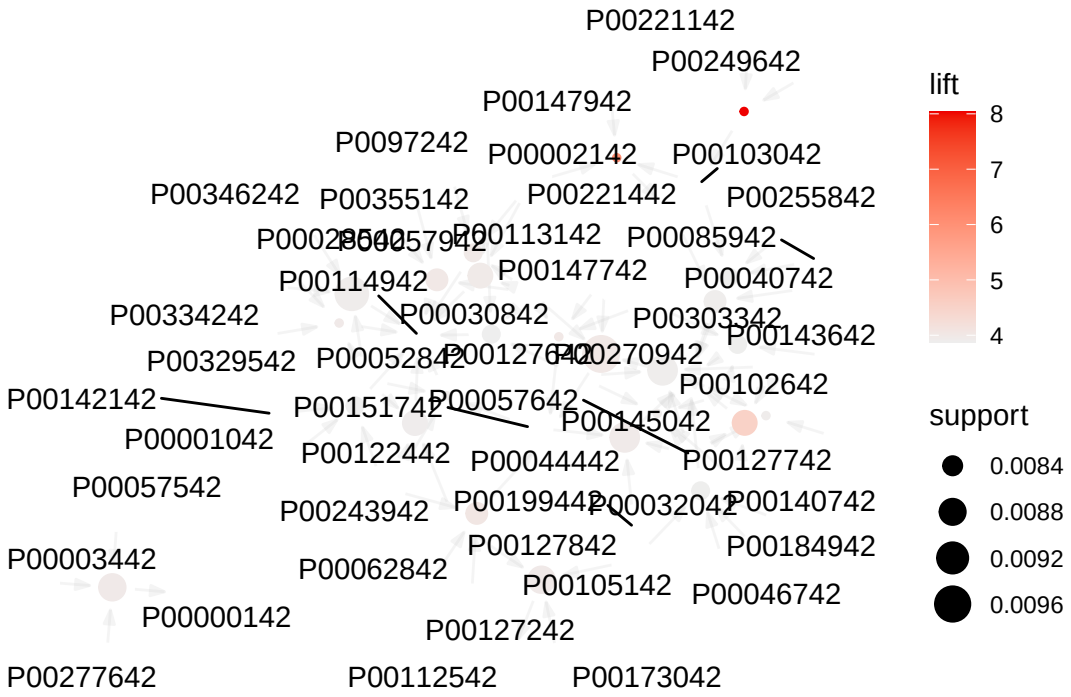
```
##      P00102642,
##      P00145042}  => {P00270942} 0.008655804  0.8793103 0.009843856 4.540663    51
## [2] {P00025442,
##      P00031042,
##      P00034742,
##      P00255842}  => {P00145042} 0.008486083  0.8064516 0.010522743 3.433246    50
## [3] {P00003242,
##      P00130742,
##      P00237542}  => {P00145042} 0.008316361  0.8032787 0.010353021 3.419738    49
## [4] {P00006942,
##      P00251242,
##      P00277642}  => {P00145042} 0.009674134  0.8028169 0.012050238 3.417773    57
## [5] {P00034042,
##      P00112442,
##      P00112542}  => {P00110742} 0.008146640  0.8135593 0.010013578 3.012880    48
## [6] {P00127642,
##      P00165442,
##      P00277442}  => {P00110742} 0.008316361  0.8032787 0.010353021 2.974807    49
## [7] {P00051442,
##      P00112142,
##      P00112542,
##      P00270942}  => {P00110742} 0.008146640  0.8000000 0.010183299 2.962665    48
```



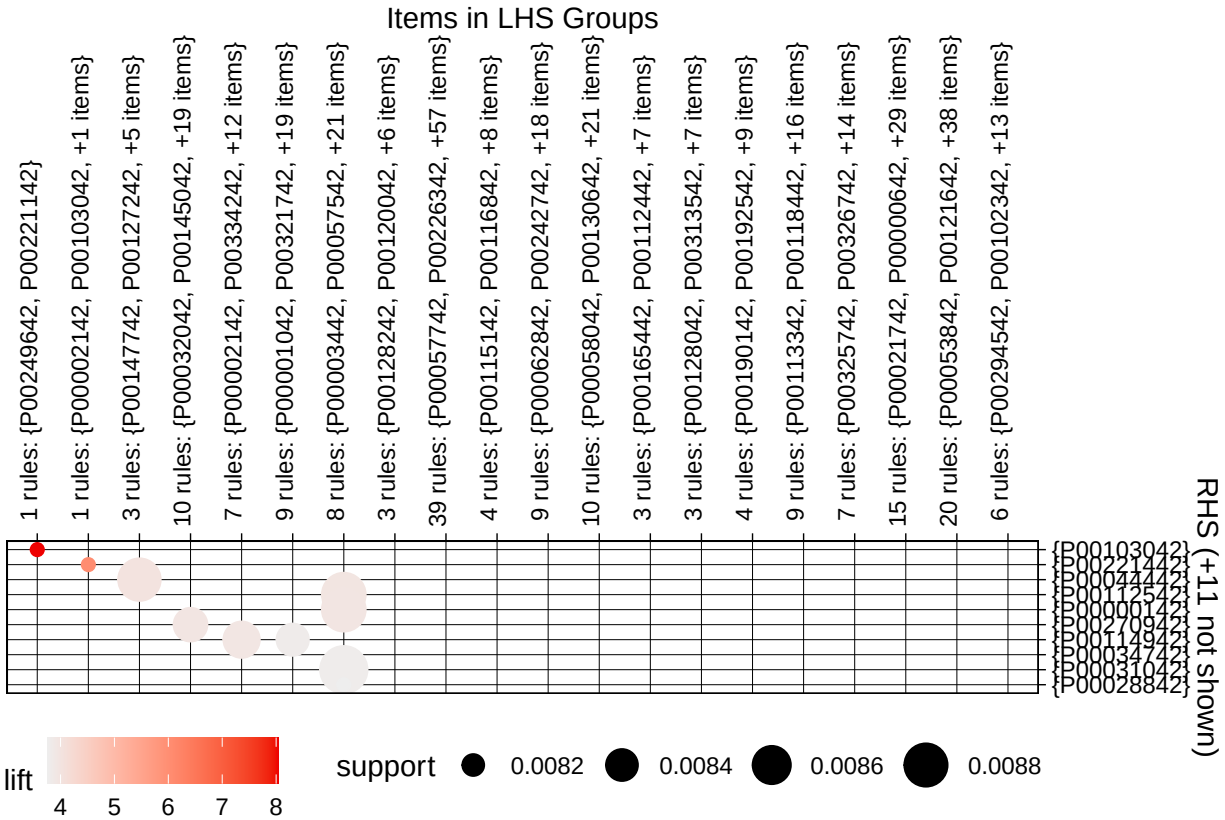

```
## Apriori
##
```

```
## Parameter specification:
##  confidence minval smax arem  aval originalSupport maxtime support minlen
##        0.75    0.1    1 none FALSE            TRUE       0   0.008      1
##  maxlen target  ext
##      10  rules TRUE
##
## Algorithmic control:
##  filter tree heap memopt load sort verbose
##     0.1 TRUE TRUE  FALSE TRUE    2    TRUE
##
## Absolute minimum support count: 47
##
## set item appearances ...[0 item(s)] done [0.00s].
## set transactions ...[10539 item(s), 5892 transaction(s)] done [0.58s].
## sorting and recoding items ... [2099 item(s)] done [0.02s].
## creating transaction tree ... done [0.01s].
## checking subsets of size 1 2 3 4 5 6 done [58.49s].
## writing ... [171 rule(s)] done [1.25s].
## creating S4 object  ... done [0.59s].

##     lhs             rhs             support confidence    coverage     lift count
## [1] {P00221142,
##      P00249642}  => {P00103042} 0.008146640  0.7619048 0.010692464 8.030667    48
## [2] {P00002142,
##      P00103042,
##      P00147942}  => {P00221442} 0.008146640  0.7500000 0.010862186 6.045144    48
## [3] {P00032042,
##      P00057642,
##      P00102642,
##      P00145042}  => {P00270942} 0.008655804  0.8793103 0.009843856 4.540663    51
## [4] {P00062842,
##      P00127242,
##      P00243942}  => {P00044442} 0.008486083  0.7575758 0.011201629 4.061544    50
## [5] {P00030842,
##      P00057942,
##      P00355142}  => {P00114942} 0.008486083  0.7936508 0.010692464 4.024260    50
## [6] {P00030842,
##      P00147742,
##      P00303342}  => {P00044442} 0.008146640  0.7500000 0.010862186 4.020928    48
```




```
## Available control parameters (with default values):
## k     =  20
## aggr.fun  =  function (x, ...)  UseMethod("mean")
## rhs_max   =  10
## lhs_label_items   =  2
## col   =  c("#EE0000FF", "#EEEEEEFF")
## groups    =  NULL
## engine    =  ggplot2
## verbose   =  FALSE
```



We can observe that the sparse matrix has dimensions of 5892 by 10539, indicating 5892 rows (also referred to as elements, itemsets, or transactions) and 10539 columns (items). By using the summary function, we are able to obtain a density of 0.008768598, which implies that our matrix contains 0.9% non-zero values (1) and 99.1% zero (0) values.

Furthermore, our Exploratory Data Analysis has revealed that the summary() function provides us with the most frequently purchased items by customers. To confirm our findings, we can cross-check this data with what we previously discovered. The resulting output from our sparse matrix is as follows:

1. P00265242 = 1858
2. P00110742 = 1591
3. P00025442 = 1586
4. P00112142 = 1539
5. P00057642 = 1430
6. (Other) = 53648

Based on the above data, it appears that our top 5 best-selling products (identified by product ID) are as follows:

1. P00265242 = 1858
2. P00110742 = 1591
3. P00025442 = 1586

4. P00112142 = 1539
5. P00057642 = 1430

The distribution of the number of items in a customer's basket, also known as the "element length distribution," provides us with valuable information. The distribution reveals the number of items purchased by customers, and we can also view additional information such as quartiles and mean values. In this instance, the mean value is 92.41, indicating that customers, on average, purchased 92.41 items. However, since some customers purchased over 1000 items, the mean value may be skewed. Therefore, it may be more appropriate to use the median value of 54.00 items, which is less influenced by outlier values.

To gain a better understanding of the items, we can create an item frequency plot using the arules package.

This visualization shows the association rules generated by the Apriori algorithm. The top section of the visualization displays the LHS (left-hand side) of the rules and the right-hand side section shows the corresponding RHS (right-hand side). The size of the bubbles in the visualization represents the support value of the rule, which indicates the frequency of occurrence of the rule in the dataset. The fill/color of the bubbles represents the lift value, which measures the strength of the association between the LHS and RHS. A higher lift value indicates a stronger association between the items in the LHS and RHS.

4 Conclusion

Our exploration of the Black Friday dataset yielded many valuable insights. We were able to gain an understanding of how our customers are classified based on demographic information like gender, age, occupation, and residency. Additionally, we identified the top customers in terms of purchasing activity, and determined which products were the most and least popular during the Black Friday sale. In terms of overall purchase metrics, we analyzed the average amount spent by customers and the total amount spent across various categories.

After completing our EDA, we delved into Association Rule Learning and discovered several rules that could be useful for our store. These rules indicate that customers who purchase certain items are highly likely to purchase another item as well, based on the patterns we observed in the data. These insights can be invaluable in understanding customer behavior and making strategic business decisions.

5 Reference

1. https://www.kaggle.com/code/dessinhaprado/black-friday/input
2. https://towardsdatascience.com/exploratory-data-analysis-8fc1cb20fd15
3. https://en.wikipedia.org/wiki/Apriori_algorithm